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Workpackage D

Smart Energy

Deliverable 3

Implementation of smart meter data in the production of official statistics

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Prepared by:

Arko Kesküla, Tõnu Raitviir (Estonian Statistics, EE)
Ingegerd Jansson (Statistics Sweden, SE)
Tatsiana Pekarskaya, Johan Fosen (Statistics Norway, NO)
Maria Rønde Holm (Statistics Denmark, DK)

Workpackage Leader:

Arko Kesküla (Estonian Statistics, EE)
arko.keskyla@stat.ee
mobile phone : +372 56673210

Abstract

The Smart Energy work package D (WP D) is one of the work packages in the ESSnet Big Data II project and aims to implement smart meter data for production of official statistics. This report is the third and final report out of three. It gives a brief overview how WP D aligns to Big Data REference Architecture and Layers (BREAL) concentrating on the information architecture. The report further covers the quality framework and risks that are involved when using smart meter data. Moreover, information and suggestions about smart meters metadata, data delivery, storage, validation, preparation and linking to administrative sources are covered. The report ends with describing methodology on how to find the electricity consumption of businesses and households, and how to identify empty dwellings.

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1 Introduction

The use of smart electricity meter data and appropriate analytical methods will enable the European Statistical System (ESS) to produce new kinds of statistics or support traditional existing statistics.

A smart electricity meter measures electricity consumption and/or production at a high frequency and communicates the information to a central system. Typically, smart meters transmit data to the electricity provider on an hourly basis. The smart meter has a location with an address that can be translated into a geographical point. Generally, a smart meter will either be of production or consumption type, but there are also combined types that measure both production and consumption.

The electricity market comprises a number of actors: network operators, electricity providers, customers and others. Many of the Nordic countries have adopted a setup where data are gathered within a central institution that manages a data hub. A data hub could look like Figure 1.

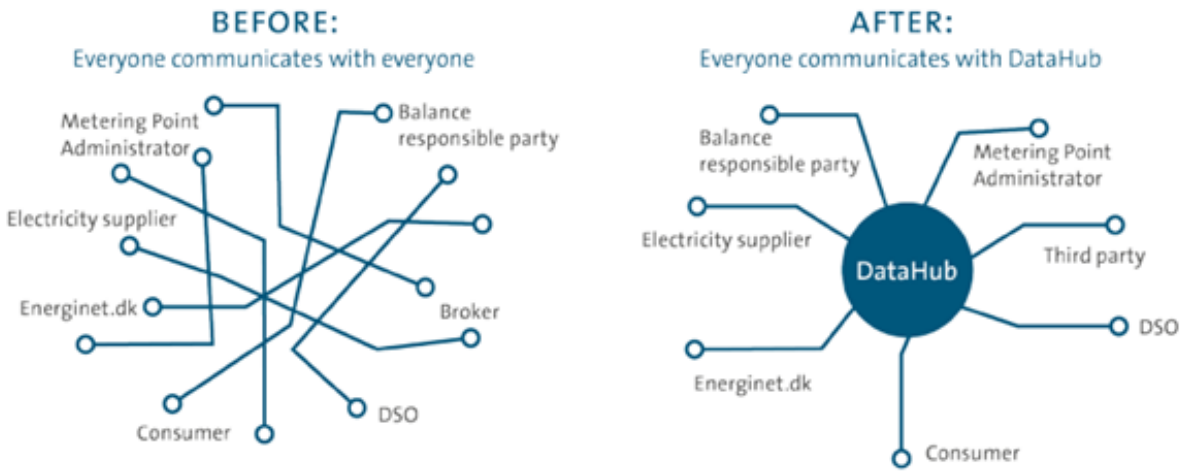


Figure 1: Example of Danish data hub

In the data hub, all data related to a metering point is collected and stored centrally. Information that is received in a data hub on smart meters, can be divided in two large groups: background data and consumption/production data. The former group contains information about:

- smart meter characteristics (smart meter identification number, energy reading type, installation address and other reading characteristics),
- end user characteristics (id, living/invoice address, contact information, subscription plan),
- electricity and greed access providers information.

The consumption/production data group contains measures and information about consumption and production volumes. Both groups of data can be associated with a timestamp, showing when a measurement or a change in the background information was done.

The National Statistics Institutes (NSIs) receive both of these data sets from the data hub.

The aim of work package D is to implement the use of smart meters data for producing statistics in different areas, e.g. energy statistics of businesses, households, census statistics on vacant dwellings. The implementation will include linking electricity data with other administrative sources for producing statistics of businesses and households, and identifying vacant living places or seasonal/temporary occupancy of living places. The duration of the project was 24 months with four participating countries (Estonia, Denmark, Norway and Sweden).

It is possible for the statistics producer to collect smart meter data directly from the electricity or grid provider, but it is a great advantage for the use of smart meter data in statistics production if the data are available through a central national data hub. Currently, Denmark, Estonia and Norway have national hubs in operation, while Sweden is planning for a hub. In this report, we assume that there exists a central data hub.

During the project, implementation procedures for the following statistical products will be produced:

- electricity statistics of businesses, by sector
- electricity statistics of households
- identifying vacant or seasonally vacant dwellings by new estimation models

This includes setting up procedures and developing technical solutions to promote and support the collection, processing, and analysis of the data for statistical production. Additionally, the national hubs will enable the participating NSIs to produce country specific statistical products, for example statistics of finer granularity, new housing statistics, improved statistics on type of production and prepared data available for researchers. Other possible benefits are lower response burden, higher quality, and faster production. One could also benefit by producing statistics on household costs, tourism seasonality, or impact on the environment.

2 Alignment to BREAL

BREAL (Big Data REference Architecture and Layers)[1] is a European reference architecture for Big Data (BD) that is being actively developed by Work Package F[2] on ESSnet Big Data 2. BREAL helps NSIs to develop standardised solutions and services to be shared within the ESS and beyond. It is particularly useful for NSIs that aim to introduce the use of Big Data in their production processes, especially those that plan to use Web or sensor data.

2.1 Information architecture for WP D

In this section, general information architecture for smart meters is described using BREAL Generic Information Architecture for Big Data (GIAB)[3]. There are three defined layers: Raw data Layer (Figure: 2), Convergence Layer (Figure: 3) and Statistical Layer (Figure: 4)[i].

The Raw data Layer contains all necessary data resources that are acquired during the Acquisition and Recording phase. Many of the Nordic countries have adopted a setup where data are gathered within a central institution, which manages a data hub. All data relating to a metering point are collected and stored centrally in the hubs. Hub data contain information on metering points, customers, agreements and the consumption/production of energy.

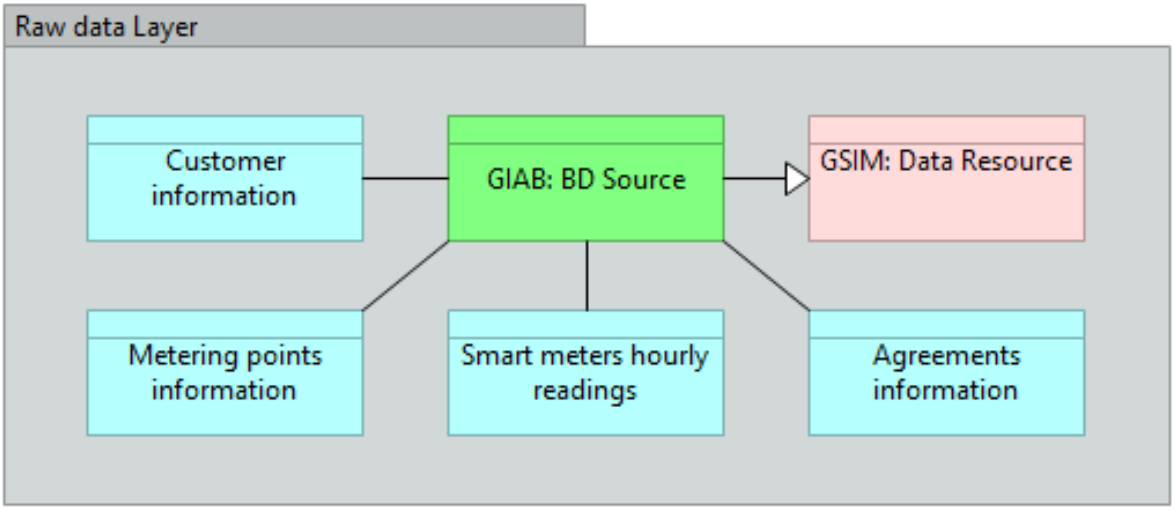


Figure 2: BREAL - Raw data Layer[i]

[i]BD - Big Data, GSIM - Generic Statistical Information Model.

The Convergence Layer contains data represented as units of interest for the analysis. Main focus objects are households, business units and dwellings. As an additional resource, business register and weather data are used. Data Representation and Data Wrangling business functions and corresponding application services are responsible for creating and moving data in this layer.

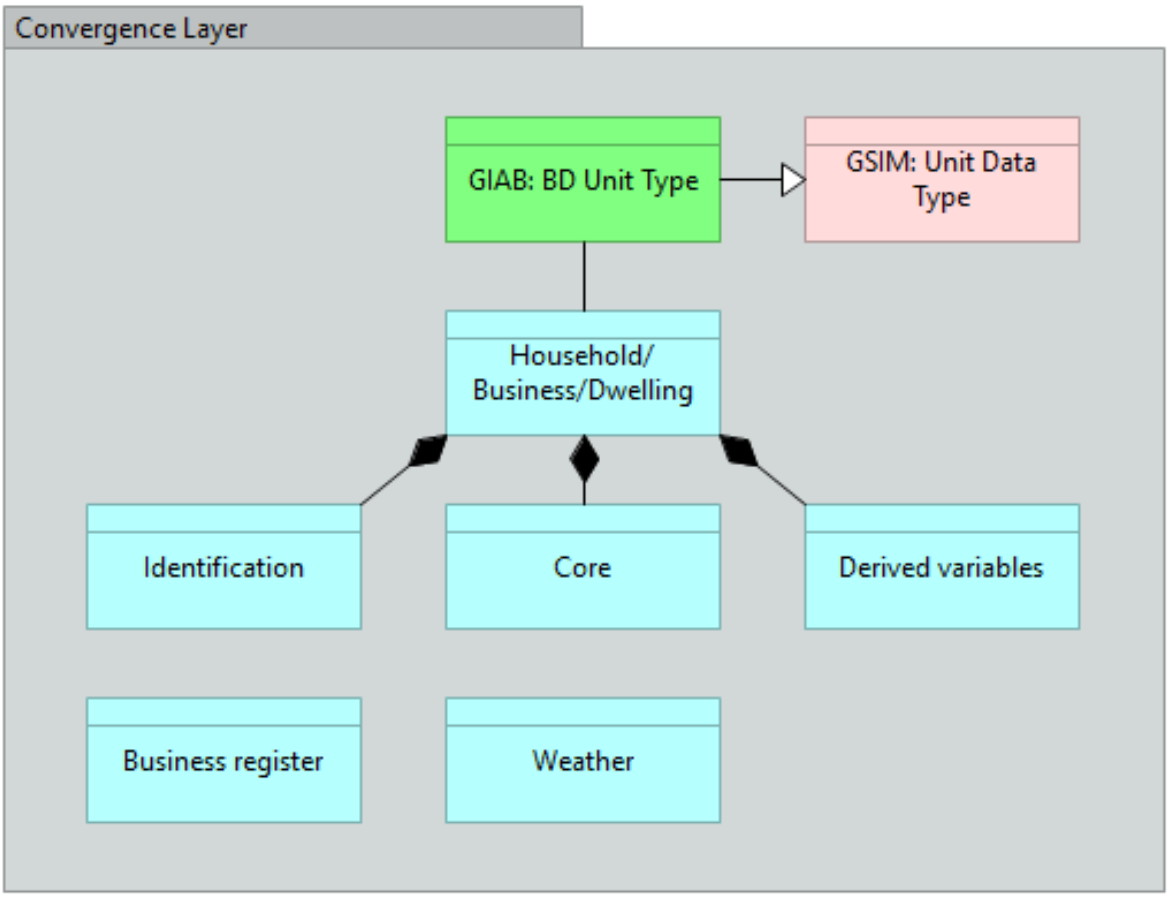


Figure 3: BREAL - Convergence Layer[i]

The Statistical Layer includes those concepts that are the targets of the analysis, which in our case are:

- Electricity statistics of businesses, by sector
- Electricity statistics of households
- Identifying vacant or seasonally vacant dwellings by new estimation models

Modelling and Interpretation and Shape Output business functions are used to operate with data in this layer.

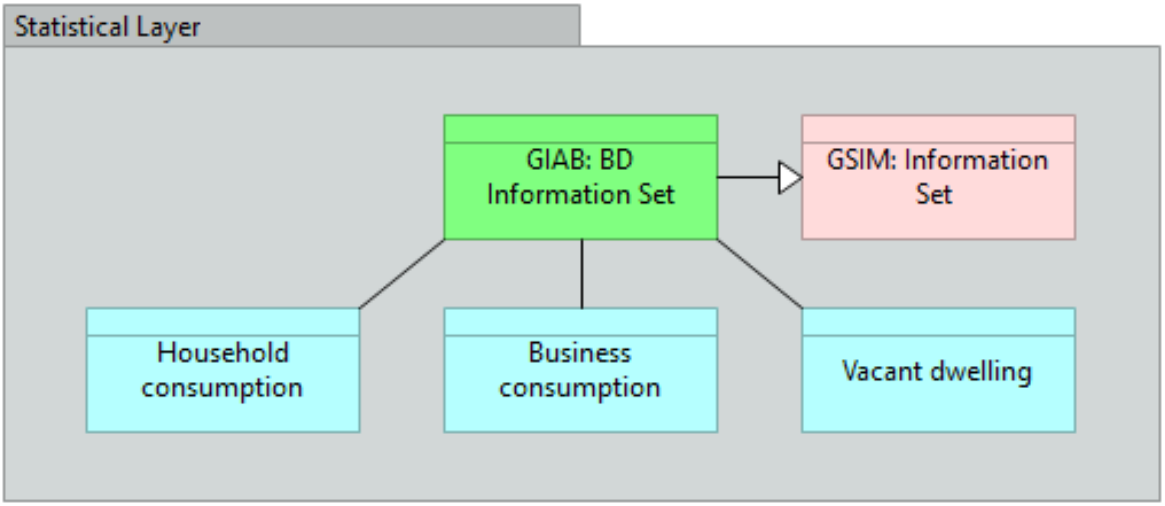


Figure 4: BREAL - Statistical Layer[i]

3 Quality framework

A quality framework should ensure the quality of both the input data, the processes for creating the statistics, and the resulting statistical output. In the previous ESSnet Big Data, the first list of quality indicators was tested on available data. The intention was to test if the indicators were good measures of quality, and if the quality of the data was satisfactory. In addition, quality indicators can be used for comparisons between countries. At the time of the previous ESSnet Big Data, Estonia and Denmark were the only participating countries with enough hub data available to calculate the indicators.

In the present project, the list of indicators is further developed, and linked to the work of Work Package K[4] in the current ESSnet project. Work Package K has issued suggested quality guidelines for the acquisition and usage of big data. They focus mainly on two phases of the production process (the input and two layers of throughput). Quality indicators for the output should not be dependent on the data source and thus the traditional quality dimensions for output hold.

New data sources differ substantially in the input phase, compared to administrative or survey data. Here, negotiation with data owners, preparation of workflows and infrastructure, and acquisition, recording and validation of the input data take place (see also section 9 in this report). Relevant indicators refer to the quality as agreed between the NSI and the data owner, such as

- Periodicity, i.e frequency of receiving the data set at the NSI
- Processing time by the data owner, i.e. time between registration and the time when the NSI receives the data.
- Delays in the delivery of data compared to the agreed delivery time
- Are the number of files and file formats correct?
- Are the number of variables, variable names, variable types, etc. correct?

At the lower level of the throughput phase, unstructured data are processed into well structured (statistical) data. At the higher level, the statistical data are used to produce the statistical output. Within this framework, possible error types are categorised as coverage, comparability over time, measurement errors, model errors, and processing errors.

The quality dimensions identified in the report of Work Package K, as relevant for smart meter data, refer to coverage and accuracy, comparability over time, linking and modelling errors, and process errors. Below, some examples of smart meter quality indicators that we identify as useful are grouped according to this classification.

Coverage and accuracy:

- Proportion of households and companies that do not have smart meters, at national and regional level
- Proportion of consumption that is not covered
- Percent of units where consumption/production is imputed

- Percent of units that fail basic checks, such as checks for outliers or negative values
- Number of outliers per metering point
- Percent of units being duplicated in the received data set
- Percent of missing readings
- At a macro level, consumption and production could be compared with the estimates from available household or business surveys

Comparability over time:

- Development of smart meter deployment over time
- Is there a long-term contract with the data owner?

As an example, Figure 5 shows the development of smart meter coverage in Denmark.

Linking and modelling error (see also subsection 8.3 about linking, classification and modelling of consumption):

- Are unique keys available?
- If not, have additional linking operations been developed (i.e using non-unique information or probabilistic record linkage).
- Quality checks after linking

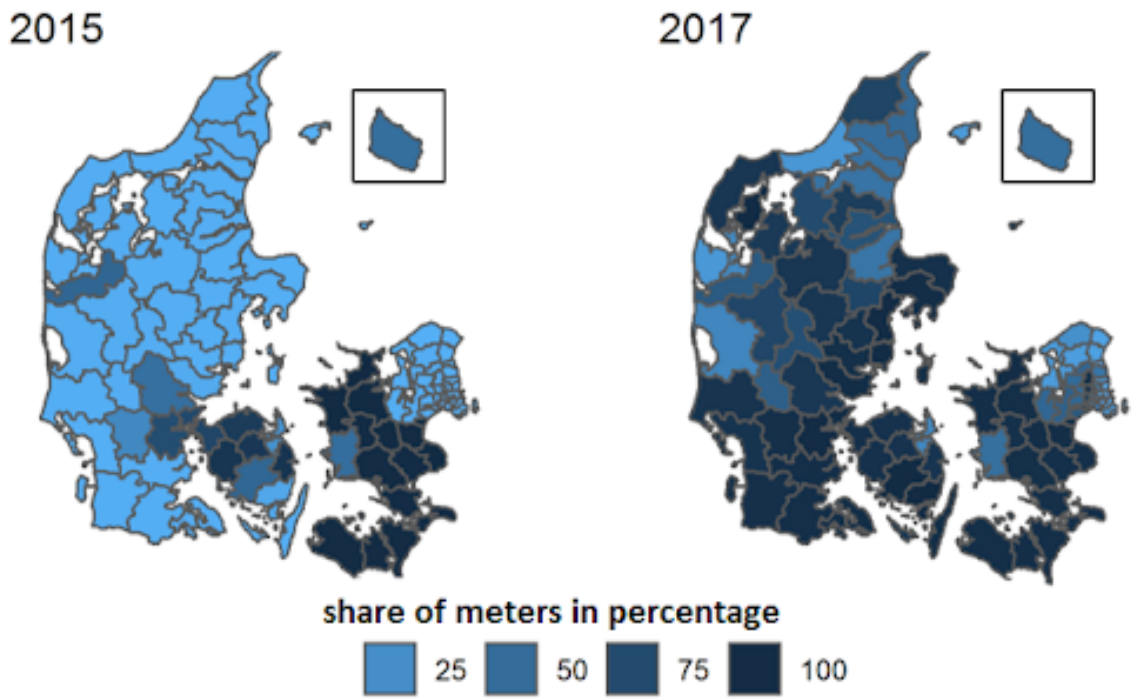


Figure 5: Development of smart meter coverage in Denmark

  - proportions of linked units
  - comparisons with an audit sample in the case of probabilistic linking

- Description of the difference between administrative and statistical units
- Checks of classification (e.g NACE or vacant/non-vacant dwellings) by an audit sample
- Checks by an audit sample when consumption is divided on businesses or households by a model

Process error:

- Continuous mentoring of production and consumption to locate unusual fluctuation in time series
- Assess own consumption of own produced energy

4 Risk plan and mitigation scenarios

The main risks that can affect the delivery of results for all participating countries in work package D can be split into two types of risks:

Data handling risks:

- Protection of personal data fails according to the requirements of the General Data Protection Regulation (GDPR) and other relevant regulations. Electricity consumption data are considered as private by many users, and the perceived threats to privacy is likely to increase if the granularity of the data is fine. As an example, electricity consumption may reveal which periods of time that a dwelling is not used. Also, although the inference will be very uncertain, the consumer might be worried that the consumption may suggest some patterns concerning time and frequency of cooking dinner, showering, etc. The NSI has access to many administrative registers and has a security policy to ensure compliance with GDPR, and the risk related to electricity data should not be different from the well-known risks related to any register. The consequences of failing to protect personal data could be large due to the perceived risk level among consumers. The consequences could be damage to public trust in the NSIs ability to protect data collected for the production of official statistics.
- Linking data to other sources fails or is not of good enough quality.
- The data provider changes the data structure in a way that affects the processes at the NSI, but fails to communicate changes to the NSI.
- Poor quality of the outputs.

Administrative risks:

- Unclear contract with the data provider.
- Failure to cooperate with the data provider.
- Changes in legal aspects opening for the data provider to refuse delivering the data.
- Lack of resources and people.
- Technological or administrative changes that make previous work obsolete.

To meet and mitigate the risks, it is important for the participating countries to set up contracts with the data providers and have good cooperation with them. We must also be able to adapt swiftly to changes in data structure and techniques, and for example investigate the use of models for estimating outputs. Also, we need to make sure that people are trained and that knowledge is transferred, to avoid losing momentum in case of changes in staff.

There are also a number of country specific risks. Denmark had promised to deliver data to researchers, but if there is no benefit for the data provider, they might refuse delivering data. Sweden does not have a hub yet and there is a risk of major changes that will affect

the possibility of using the data in the way currently intended. The country specific risks will not be handled by the project.

Data should be treated with care, as they are sensitive personal data. The local GDPR officer should be involved in retrieving data from the data provider. The contract should include a section on data security. To make the data collection smooth, laws should be in place so that they protect the individuals but allow the use of data for statistical and research purposes. Typically, a few people are involved in the preparation, collection and assembling of data. Afterwards, when data are ready to be used in statistical production, more people will need to have access to data. Generally, the same rules as those for any sensitive data should apply.

5 Metadata

Metadata contain information available through the hub, as well as additional information linked from other administrative sources.

Data from the data provider contain information on metering points, customers, agreements and the consumption/production of energy (see Table 1)

Table 1: Information on metering points, customers, agreements and the consumption/production of energy

Variables group	Variables list
Time	• Date or/and time for which information in a row is valid. Detalization of the variable depends on the aggregation level of information in the row.
Smart meter characteristics	• Metering point id • Metering point status (active, inactive, terminated) • Metering point type (production, consumption, combined or other). • Meter reading mode (automated, manual, not metered) • Metering point physical address (street, street number, house number, postal code and optional dwelling number, GPS coordinates of the metering point). The address information can be provided in several formats depending on standards used in a country.
End user characteristics	• End user id. For businesses it is organisation number, for persons - person identification number. • End user invoice address (street, street number, house number, postal code and optional dwelling number). Not all countries receive it. If received, the address information can be provided in several formats depending on standards used in a country. • In some cases End user name for companies/businesses • Other information relevant to End user
Information on meter readings	• Consumption/production data • Unit of measure (kWh) • Production type (thermal, wind, hydro, nuclear, solar or other) • Consumption type (consumption, loses, pumped, etc.) • Other information related to meter readings
Subscription information	• Smart meter NACE code: showing purpose of use (household, cabin, street light, green house, NACE for companies/organisations, or other) • Grid Access Provider and Balance supplier info • Other information related to the subscription.

While receiving the data a description of available variables should be provided by the data provider.

Linking smart meter hub data with other data sources (usually administrative registers such as business-, population- and/or of buildings- or dwellings register) leads to an ex-

tension of available information, while the correspondent metadata will vary depending on which data sources are used.

5.1 Address information

Smart meter physical address and end user invoice address (described in Figure 1) can be grouped as smart meter hub addresses. In addition, two other types of addresses might be defined: electricity address (the address of a utility unit which is served by a smart meter) and end user registered address (the address of end users that are registered in the population or business registers). The two latter does not necessarily have to be specified in each country, since in most countries the smart meter physical address is equal to the electricity address.

6 Data delivery process

6.1 Data exchange protocol

The most important part of sharing data is to set up a protocol that ensures smooth data integration. This part of the process deals with the physical transaction and integration of data. The physical means of data transaction can be done through a "File Transfer Protocol" (FTP) server or actual hard drives or similar methods.

It is suggested to set up a contract with the data provider and agree on a fixed delivery frequency and fixed filenames and data types. This is important for setting up a system that automatically reads files, for instance, in a comma-separated value format (CSV) from the server into the databases. The obvious and recommended choice of data exchange is a secure FTP server. This is doable if one of the parties have an FTP-server available within its business. In most cases that party would be the statistical institute as this is the party interested in receiving data, not only on electrical meters, but also from other sources. When the NSI is the owner and user of the FTP server, the data provider logs on to the server and places the data here. The data provider can do this because the NSI has created an account for them to use on the NSIs FTP server. With one of the parties having an FTP, files can be either pushed or pulled to the receiver.

In order to successfully exchange data, a data provider contract should be in place. In the contract, the parties should define and agree on the following elements:

- **What to deliver:**
  - The number of files. Typically data consists of consumption/production data and background data. It should be agreed on how many data tables are delivered.
  - File formats and character set.
  - Aggregation level of consumption/production data (by time and geography).
  - Which variables should be delivered in the background file(s)?
  - Each file should contain agreed variables names, content, types and formats.
  - Is background data a full extract of data for every delivery or will it only be so-called delta files? Delta files here being files with new information.
  - Does the data provider deliver only consumption meters data or also production and exchange meters data? Is a metering point the same as one meter or a group of meters?
  - How should aggregated readings in the consumption data set(s) be interpreted? Is there a timestamp available in the data set(s)? If not, which period do the data refer to?
- **When to deliver:**
  - The frequency with which to deliver data.
  - What is the data relevance period (data from which period to send)?

- **How to deliver:**
  - The exact data delivery solution (FTP, hard drive, application programming interface (API)).
  - If files are compressed the compression method should be specified.
  - Who will be responsible for backups and when an error occurs?

A possible scenario is delivering zipped CSV files to a secure FTP server. For example, Denmark has agreed with their data provider that files are delivered in compression level 8 and that the character set to be used is Unicode Transformation Format 8 (UTF-8). It is advisable that it be stated very detailed in the contract how data are delivered. The contract should be formed with assistance from people from the IT department of the NSI and the data provider.

6.2 Files structure

Data from smart meter hub are typically delivered in at least three files:

1. Consumption/production data, containing the consumption/production measurements.
2. Background information on the end user, containing the ID and invoice address of the end user as well as the date of start and stop for when this end user was attached to the meter.
3. Background information on the meter, containing: smart meter physical address, type of smart meter, meter reading mode and other.

Each file contains smart meter ID and time showing when measurements (in the first file) or updates in background information (the second and the third files) took place. Since background information does not change that often, there is no necessity for the second and the third file to contain the information at the same level of time granularity as in the first file. Therefore, in order to save storage space, it is reasonable to receive only updates of background information. The end user information usually changes when a household moves in or out of a utility unit, whereas the meter data change even more rarely.

7 Data storage

We have investigated whether to use a relational database management system (RDBMS) or Hadoop (HDFS) for data storage.

Smart-meter data in relational databases is feasible and works well, but requires that you adopt an append-only life-cycle of data. All data are appended to existing tables. PostgreSQL (from now on referred to by its common abbreviation Postgres) and Oracle Database are both mature relational database systems (RDBMS) that could be used for data storage. They both feature:

- Extensive support for standard SQL.
- Features relevant for large amounts of data, including table partitioning and parallel query.
- They support a wide range of client connections, including R, Python, .Net, Java and more.
- They are multi-platform, run on Linux, Windows and varying other operating systems.

The key difference between ORACLE and PostgreSQL databases are licensing fees. Postgres is open-source licensed by an entirely free license: use it as you please, no strings attached, whereas Oracle Database has a very expensive proprietary license. Also parallel query is only available in Oracle Enterprise Edition

Hadoop by itself is not a database, but a collection of open-source software that runs as a distributed storage framework (HDFS) to manage very large data sets. Its primary purpose is the storage, management, and delivery of data for analytical purposes. At the core Hadoop is a file system (Hadoop Distributed File System) that needs additional software (for example, Impala, YARN, Hive or Spark) to analyse smart meter data.

Difference between Postgres, Oracle and Hadoop or any other relational database does not necessarily prove that one is better than the other. They all have their strong sides and weaknesses. Choosing the tools for data storage is strongly dependent on data size and the needs and requirements of the NSI. A case study with a more detailed overview and suggestions which technology to use can be found in Appendix A A.

8 Data preparation

After data have been collected, data preparation has to be done before they can be put to use. First, transformation of CSV files to the format used in the statistical office is needed. It is followed by establishing new variables needed for further processing and linking to registers.

8.1 Data anonymization

Background data sets might contain names and national identification numbers (NIN), in some cases also numbers of passports or other documents. During data preparation, all names and document numbers are removed and NINs are replaced with a pseudonymous ID number. Pseudonymous ID allows linking smart meter data with other data sources within the NSI without directly identifying the person.

8.2 Geocoding

From the smart meter physical address information, it is possible to extract the variables that are related to forming an address code: name of the street, building number, floor number, postal code, etc. The address code, when linked to correspondent GPS coordinates, is insensitive to changes in the street name over time. Depending on the level of smart meter physical address detailing, the coordinates can then be used to link the meter to the following:

- A piece of land
- A building
- A unit within a building

Irrespective of how many different types of addresses the NSI is receiving, a preferred solution is to apply the geocoding procedure to all addresses.

8.3 Linking

In all the participating countries, smart meter data are linked to administrative registers - business register, population register, or register of real estates, buildings and addresses.

One reason for linking smart meter data to the business register is, to capture the NACE (Nomenclature of Economic Activities) code for local units, so that it will be possible to get the energy consumption by economic activity. Linking to the population register might be needed to find out characteristics of the household the smart meter is associated with.

In principle linking two data sets is easy, when there is a common unique identifier that can be used to join two or more tables. The unique identifier can be a number or code that uniquely identifies a person, business or other organisation, address or some other unique object that is used by the NSIs. Linkage of two tables is only possible when the linkage variable exists in both tables. In situations where linkage is possible, problems

arise when the quality of the keys is poor or unique codes do not exist in both data sets. Then different approaches and strategies must be combined to increase the linking quality before linking is used.

The linking of a smart meter hub to registers is on the first step done based on the address code. Further, since not all smart meter physical addresses are defined up to the utility unit they are serving (electricity address) or up to the building and land area of the electricity address, one needs different approaches that would increase the linkability. For instance, one could use an invoice address and end user registered address to find more associations of smart meters and their electricity addresses on utility unit level.

8.4 Classification of smart meters

For classifying smart meters as either "serving businesses" or "serving households", end user ID together with smart meter address information should be used. For most of the cases, end user ID helps to identify whether the smart meter is used for business or household purposes. However, there are examples where e.g. the end user ID is specified as a business, but the business only owns some apartments that are rented out to their employees. Thus, associating such smart meters with business would be wrong in this case. Using address information and/or geo location is then needed. Additionally one can use the NACE code if specified in the smart meter hub, since it indicates the purpose of the use.

8.5 Modelling of consumption/production measures

Consumption/production measures need to be modified according to the desired statistical outputs, e.g. households versus business, regions, industry classification code, dwellings versus leisure utility units. In this step of the collection, the statistical office must set up a system that brings data into the desired aggregation level. After this step, one can e.g. get monthly aggregated data by metering point or monthly aggregated data by region and industry classification code.

9 Data validation

It is meaningful to validate the data at several stages of the process, from data delivery to the NSI until the final output used for publication of official statistics. In this section, data validation is split into two major steps: validation during data transfer and validation during data processing. Because of the vast amount of smart meter data, the algorithms used for data validation need to be fast and efficient.

9.1 Data validation during transfer

Upon delivery of data, the following validation provisions should be implemented. These provisions concern validating whether or not the metadata are delivered as agreed with the data provider. Thus, this is more about data formats and data size than the value of certain variables themselves.

- Do the files fulfil the requirements written in the contract?
- Do the filenames follow what was agreed upon in the contract?
- Do the variables have the expected formats?
- Are the columns separated correctly? This has a huge impact on how smoothly data are loaded into the tables in the database.
- Are there any new variables?
- Is the number of rows changing drastically from delivery to delivery?
- Are there duplicate rows?

When loading data into the tables, a log must be set up to monitor the number of rows in the received files and the number of inserted rows.

9.2 Data validation during processing

In this step, when we start working with data, it is possible to detect data irregularities related to the values of certain variables.

Below, the validation steps are described in chronological order.

Step 1, before linking with registers:

- Are there unusual fluctuations occurring in the time series. It is a good idea to make daily aggregations of data. Such time series can be used to detect unusual days in the aggregated data set.
- Are metering points from the consumption data set included in the it background data set and vica versa. Background data is the table that includes information about the address and the subscription.

- How many observations contain a zero value? How many observations are null?

Step 2 after linking with registers:

- Is it meaningful that a unit (household or business) has a metering point that measures production but not a metering point that measures consumption?
- How much of the total consumption do the households account for vs. what is the share of consumption accounted for by businesses? How large is the proportion of consumption that is not accounted for?
- If a meter is classified as belonging to a household, does the actual consumption look like the consumption you would expect from a household?

10 Data models

To fully understand the relationships and the constraints of smart meters data, Data models (DMs) are needed. DMs define the data elements and the relationships between the data elements. Data models are used to show how data are stored, connected, accessed and updated in the database management system. The DM emphasises on what data are needed and how they should be organised instead of what operations will be performed on data.

Usually the NSIs receive the data from the data hub as consumption/production and background data sets. A simple DM for smart meters data is depicted in (Figure 6) where the consumption/production table is labelled as metering data where information about the amount of produced and consumed electricity is stored. The background data set is stored in three different tables which contain information about by whom and where electricity was consumed:

- metering_points - information about location and type of metering point (possible types are: remotely readable, single and dual tariff manually readable),
- agreements - information on when electricity contract was signed/ended and type of contract,
- customers - information about private and legal persons who signed the contract

The consumption/production table has a primary key through which all background data tables are accessible. How all of the tables are connected to each other is depicted by the Information Engineering (IE) notation scheme. It is suggested that the database is normalised to a second or third normal form[5] which will help to reduce data redundancy and improve data integrity.

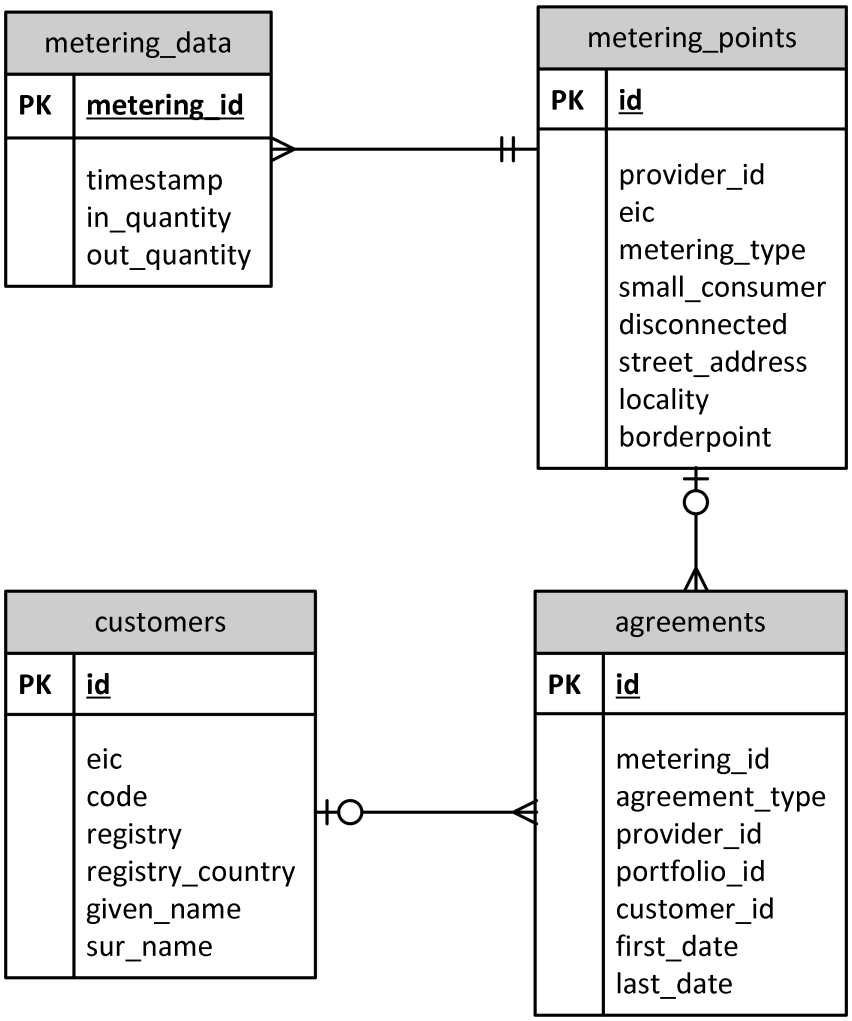


Figure 6: Example data model for smart meters

11 Methodology and implementation

11.1 Business consumption statistics

The expected outputs for business statistics are final energy consumption statistics of businesses by economic activity (monthly, quarterly and annual estimates). The overall process of finding electricity consumption of businesses can be divided into four larger phases. The first phase is Data reception and preparation, where the NSIs receive the data from the data- hub/provider. This phase includes preliminary data validation, anonymization of the data (personal codes, customer codes, etc.) and geocoding of the data. Generally a data architect is responsible for these operations with the data. An example of the Estonian workflow for this phase can be seen on image (Figure 7) where:

- Transferring the data - smart meters data sets are received and copied from the data hub or electricity provider. When all necessary files are in the production environment of the NSI, the data are loaded into a relational database (RDBMS) (Figure 7-1).
- Data are being pseudonymized - replacing personal info from customers and metering_points table with NSI specific identifiers (Figure 7-2).
- Geocoding of the data – adding NSI specific address information field to database (Figure 7-3).
- Data to BigData server - the transformed data are copied from a RDBMS database to a dedicated BigData server (Figure 7-4).

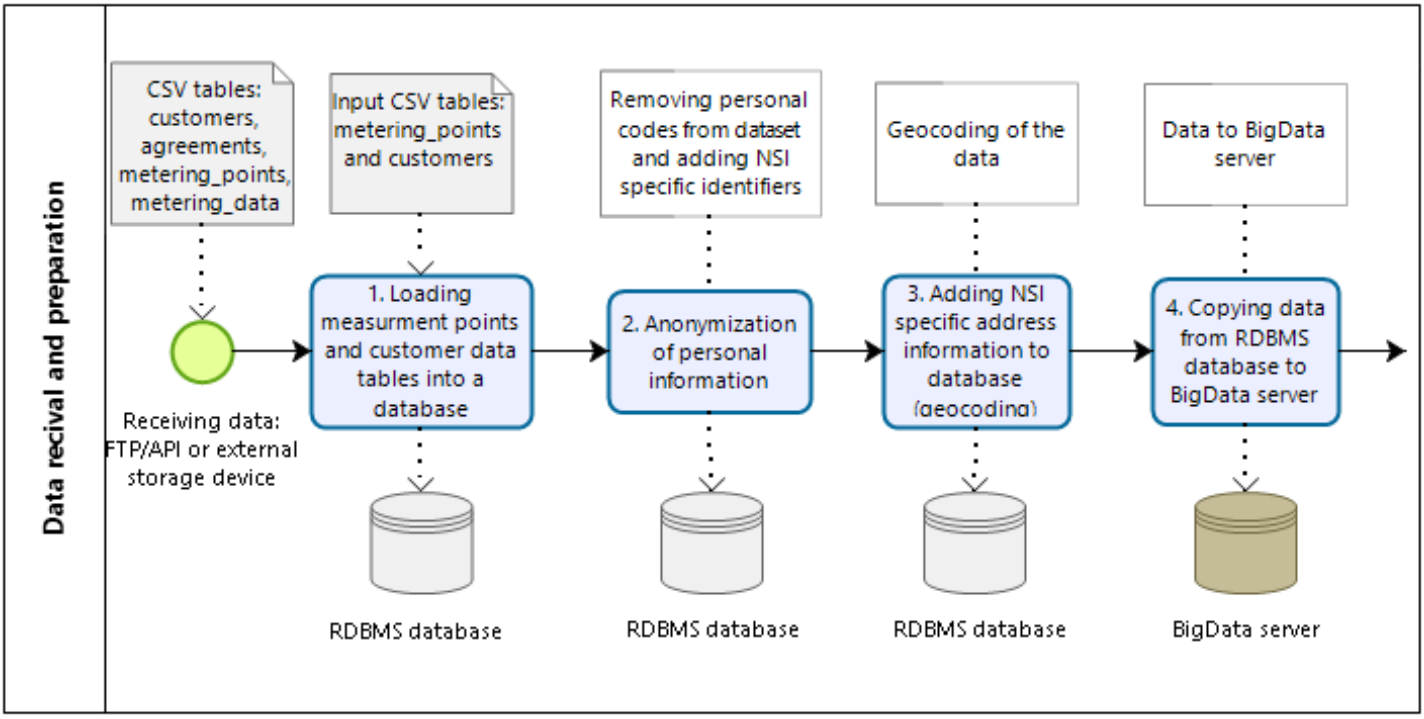


Figure 7: Data reception and preparation phase

The second stage takes place on a BigData server where the data are stored and aggregated (Figure 8):

- Data storage - The data can either be stored in a classical RDBMS (Figure 8-8) or a more big data oriented solution like Hadoop HDFS (Figure 8-5). Accessing data from distributed storage needs dedicated software like Apache Hive (Figure 8-6). Hive data warehouse facilitates reading, writing, and managing large data sets using SQL. Converting smart meters data into optimised row columnar (ORC) format (Figure 8-7) makes data access in Hive faster.

- Data aggregation - to find the electricity consumption of businesses the data granularity does not have to be very fine. Daily or even monthly aggregates will save computing power and time (Figure 8-9).

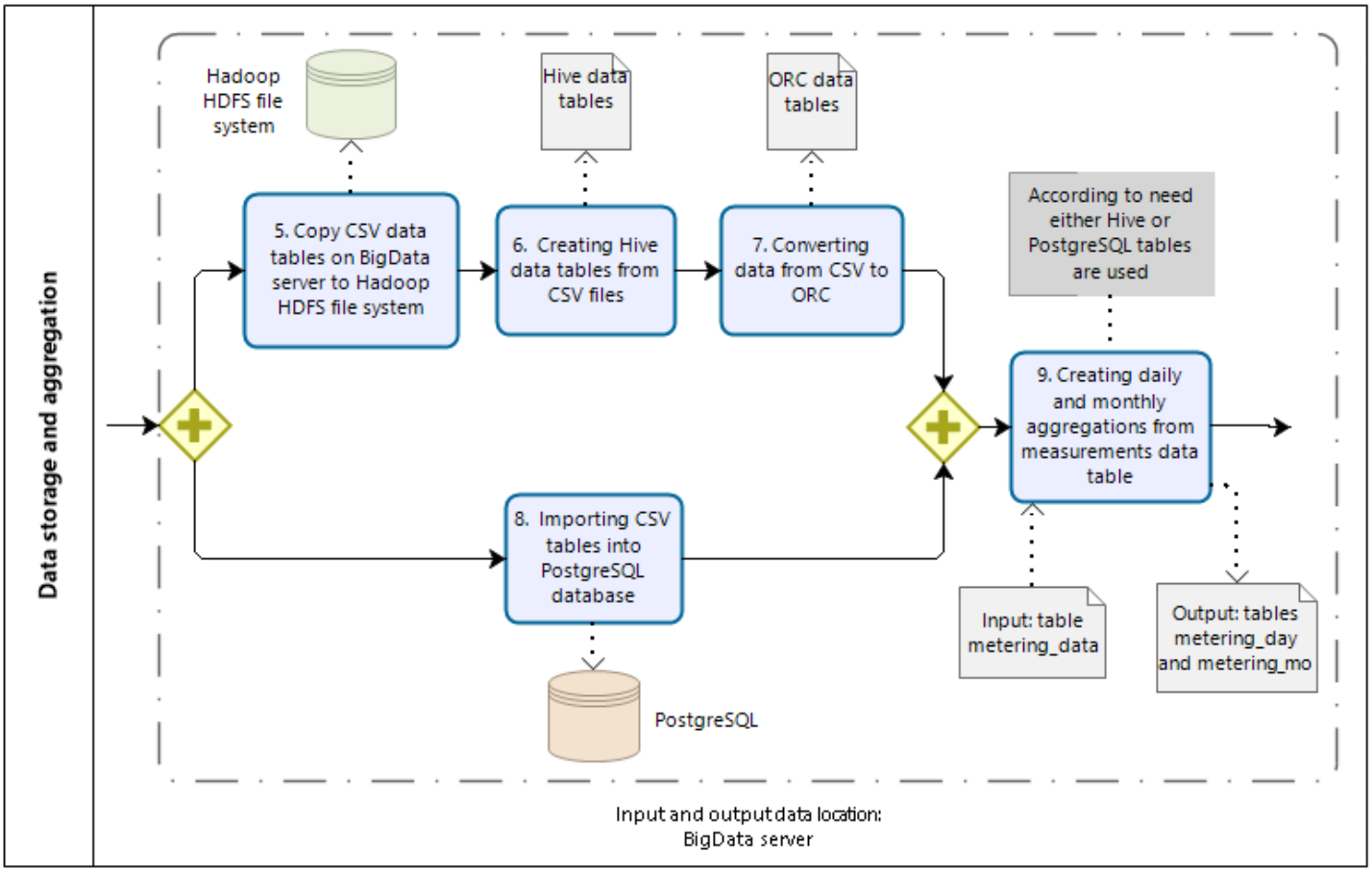


Figure 8: Data storage and aggregation

The third phase of the process is linking the electricity data to the business register. This step will enhance the data quality and give the possibility to calculate the electricity consumption statistics by economic activity sector. It is necessary to find a link between statistical units: business entities and the observed units which are metering points.

To find the statistical unit (consumption of business entity) a combined linking method is used:

- linking business customers of the data hub to business register (Figure 9-10.1) by using registry key and identifying energy consumption.

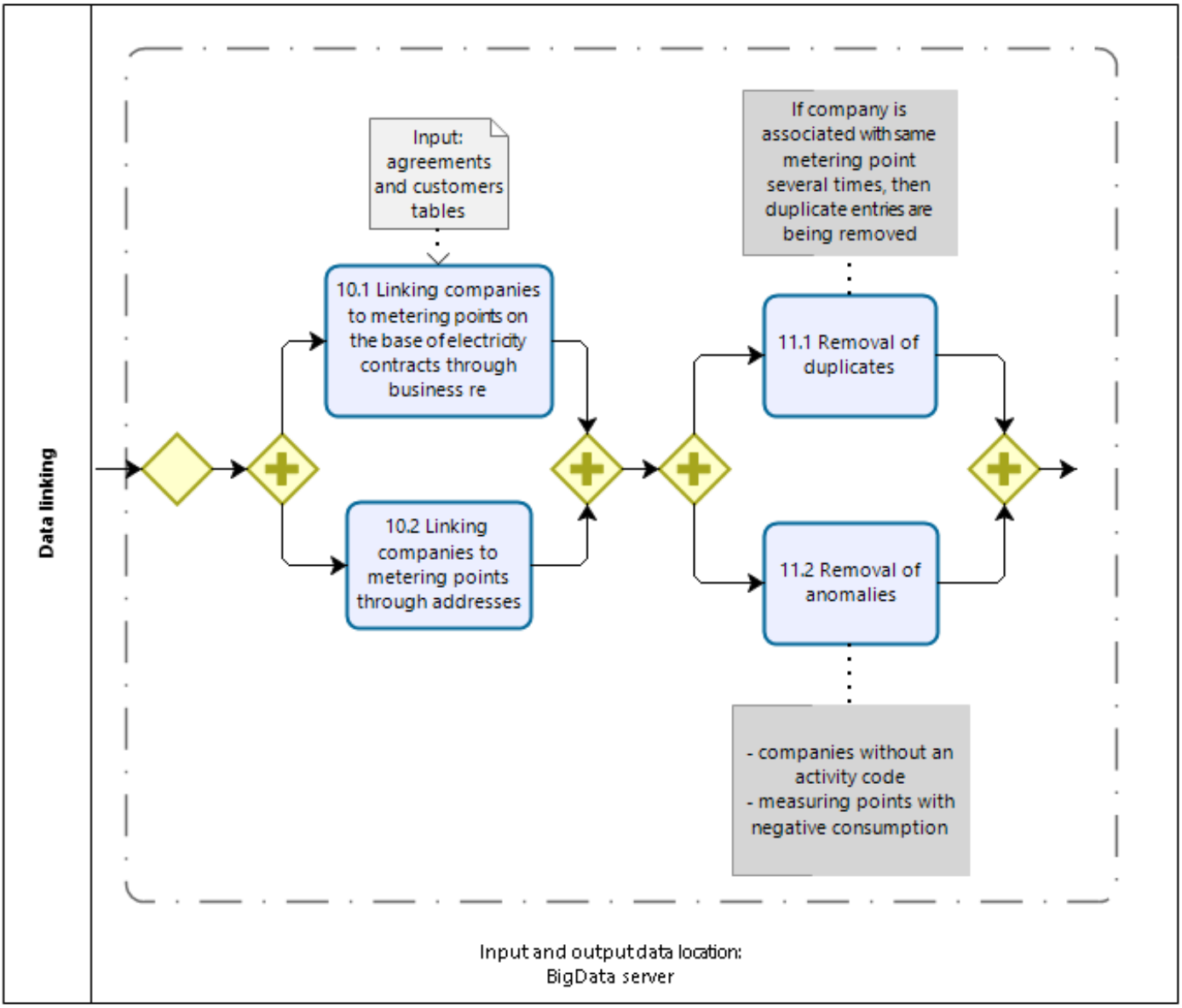


Figure 9: Data linking

- Linking the address of the business entity with the address of the metering point (Figure 9-10.2) for an estimation of consumption.

Linking is done only for metering points with a valid grid agreement. We also exclude metering points related to apartment associations, open suppliers and other network companies due to the fact that they are not the end consumers. It was not possible to exclude all the open suppliers from the further analysis as open suppliers were active in many fields and their own consumption was significant. After the linking, the data sets are combined and duplicates and anomalies are removed (Figure 9 - 11.1 and 9 - 11.2). For example if a company is associated with the same metering point several times, then duplicate entries are removed. Anomalies like companies without an activity code or measuring points with negative consumption are also removed.

In the fourth phase (Figure 10), data modelling and interpretation are performed. In this stage, problems arise when there are several statistical units (companies) on the same

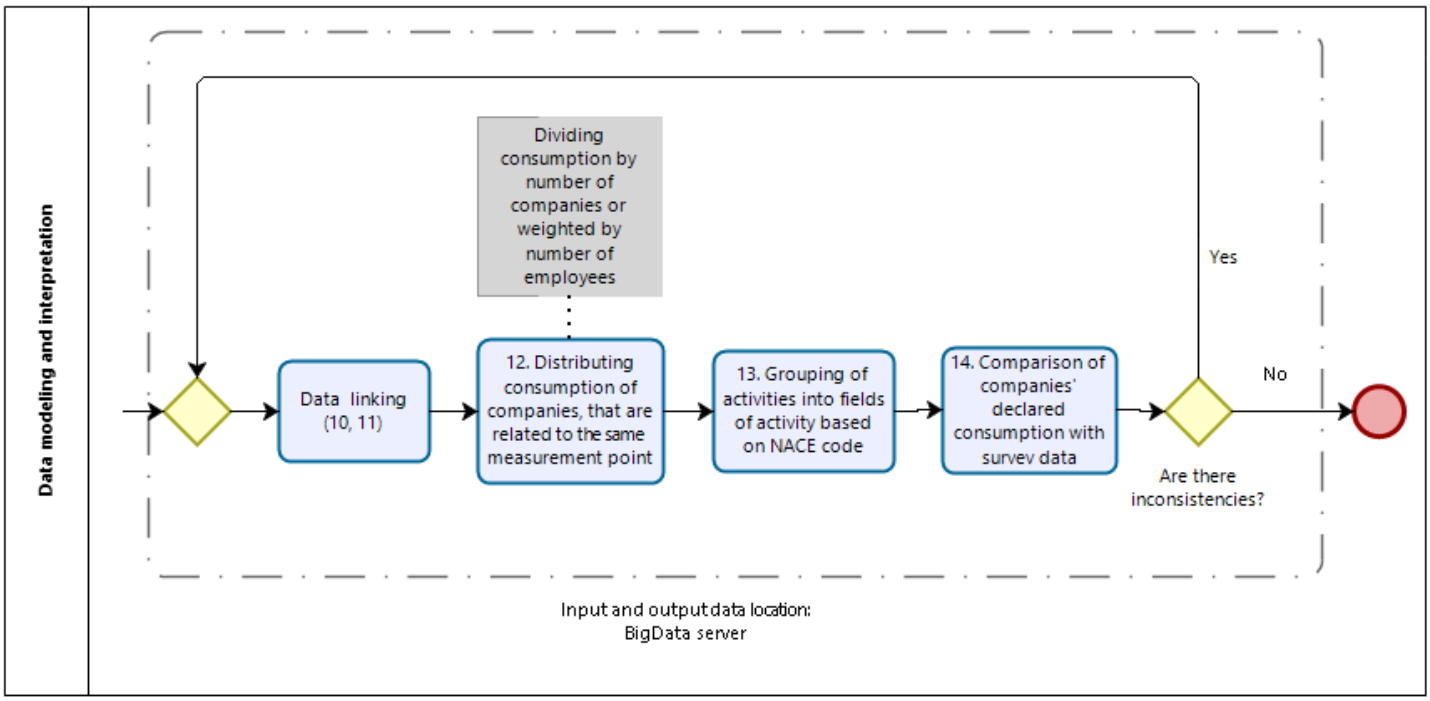


Figure 10: Data modeling and interpretation

smart meter physical address. Finding the actual end user for this metering point and identifying the amount of electricity consumed or produced might then be problematic.

Two different strategies were tested to solve the problem. Firstly the consumption is divided evenly between companies that share the same smart meter physical address. Secondly the consumption is weighted by the number of employees in the company (Figure 10-12). Alternatively the economic activity of a business can be taken into account. Choice of variables for weighting depends on the data available.

To compute the annual and quarterly electricity consumption statistics by economic activity sector, the NACE code is used to group each business into an economic sector and the consumption data is aggregated on groups (Figure 10-13).

To validate the business consumption outputs, survey data, where companies have declared their consumption, can be used (Figure 10-14). If needed the strategies of how to divide the consumption between entities can be adjusted.

Estonia's main goal was to find the total business electricity consumption by economic activity (Figure 11). The main differences between the declared consumption via survey and smart meters estimates are due to undercoverage. One explanation is that the end user's own produced electricity for their own consumption is not recorded in the data hub. We know that there are over 700 businesses that have produced electricity for the network, so we can assume that they also produce electricity for their own consumption. We found no major issues with overcoverage.

Compared to households where electricity consumption spikes during the winter months (Figure 12a) Estonian businesses (Figure 12b) have no real seasonal consumption patterns.

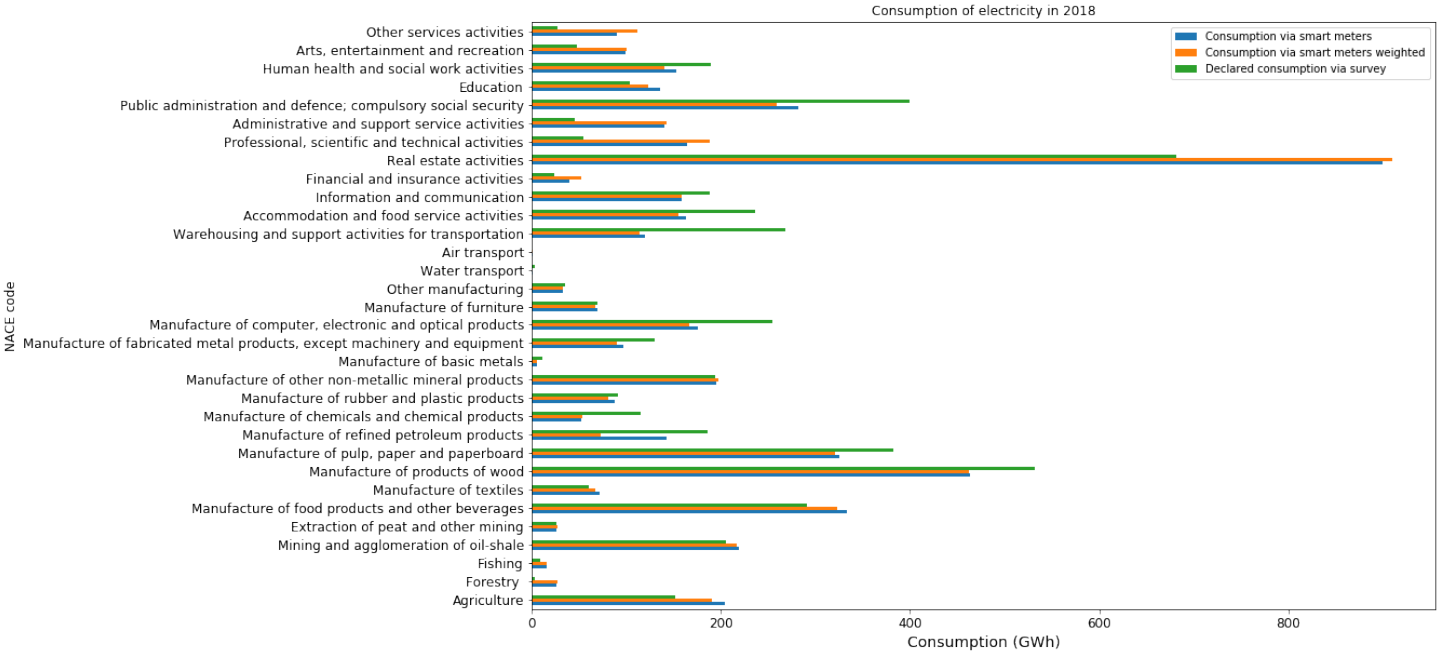
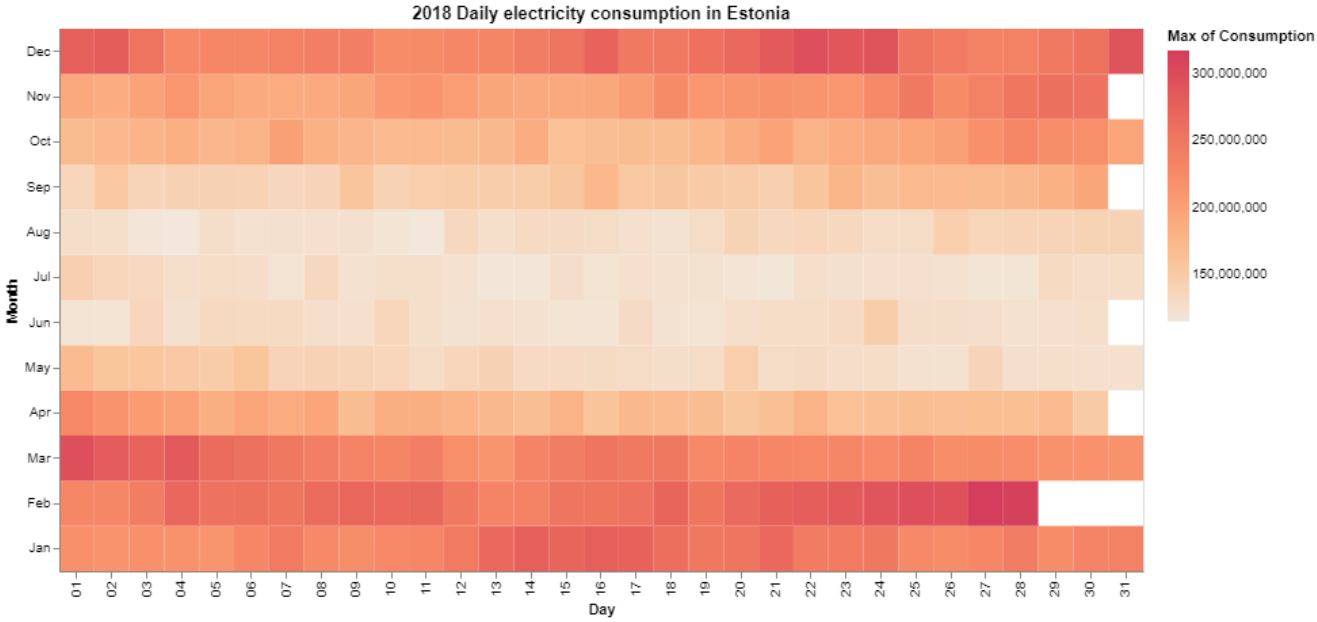
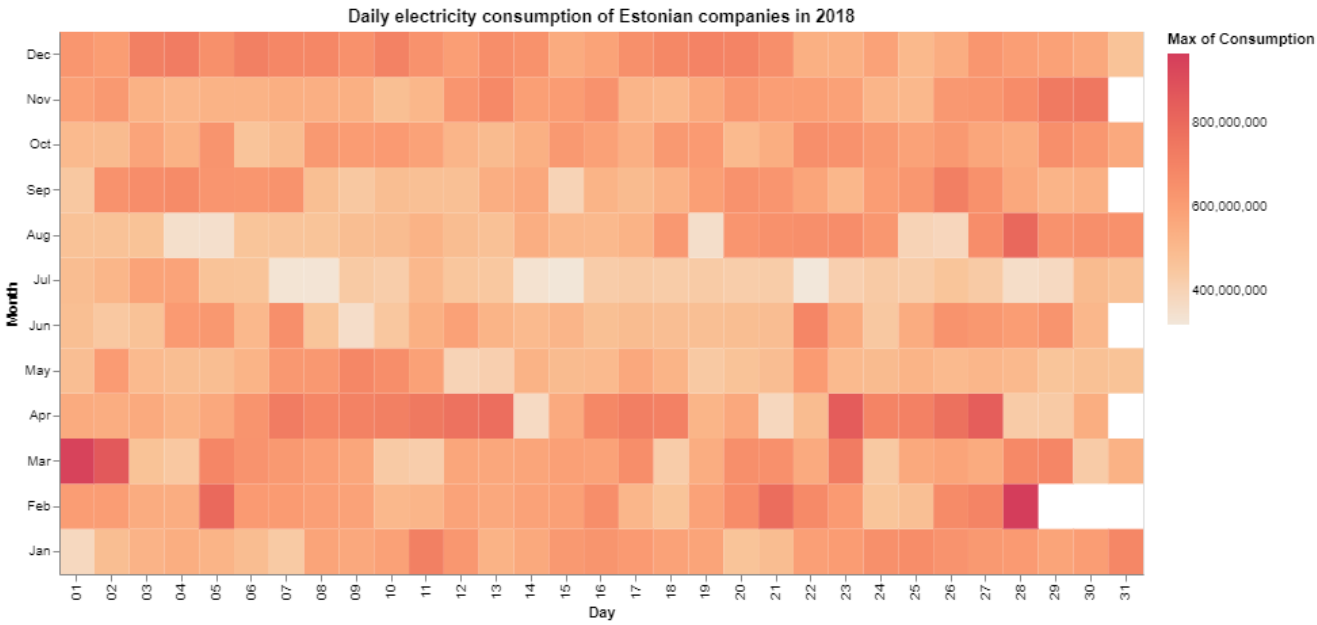


Figure 11: Electricity consumption of Estonian businesses by economic sector



(a)



(b)

Figure 12: Daily consumption of Estonian households (a) and businesses (b)

11.2 Household statistics

In this section, the Danish household statistics are presented to showcase the potential of the data. Data are received through an FTP server. They come in two types of files: 1: background data and 2: consumption data. Data are then loaded from the CSV files into the Postgres database, as shown in Figure 7.

This step is followed by a linking procedure depicted in Figure 9. Denmark receives only one address, the physical address of the metering point. This address is converted into an address id and coordinates which are linked to buildings and units inside buildings. The unit- and building codes are then used to link to the national building and dwelling register, which contains usecodes for all buildings, units and dwellings. Usecodes selected were household, summerhouse or apartment. Further, the building- and unitcodes were linked to the population register in order to find only the units where people were registered with an official address. Addresses are selected for the statistics if more than zero and less than ten people were registered on the address in 2019 .

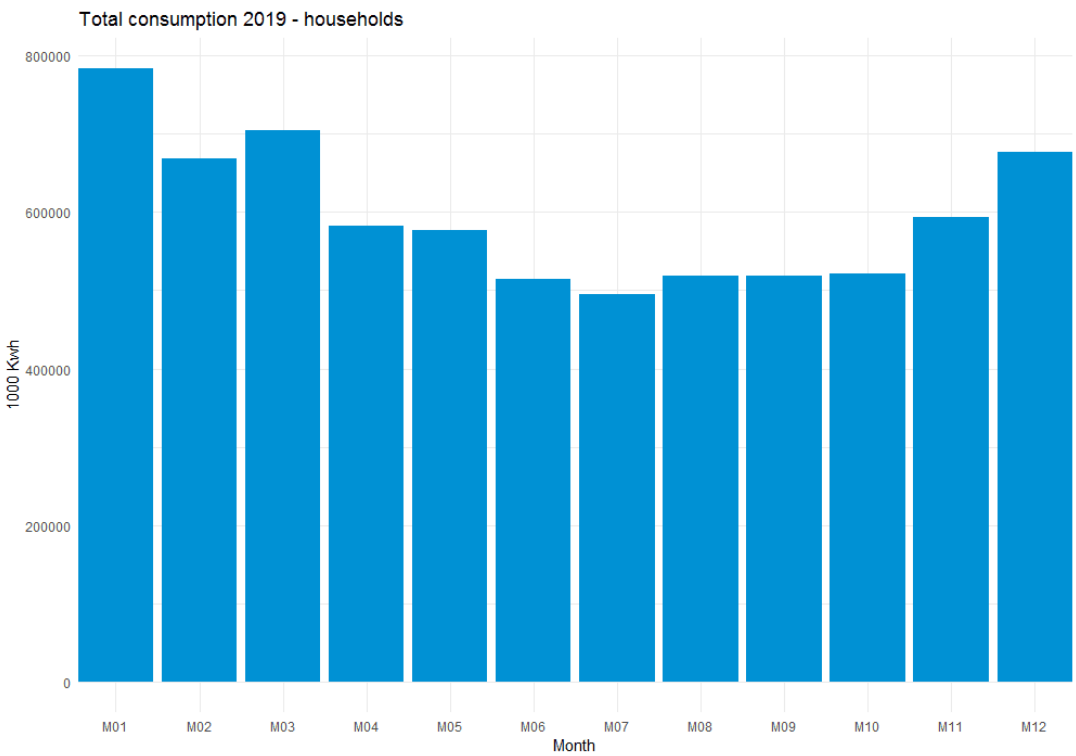


Figure 13: Monthly consumption of Danish households

In Figure 13, the total household consumption in 2019 is shown month by month. It is clear that the monthly consumption falls during the summer. In Figure 14 the average monthly consumption is shown for households with 1-2 persons in the household and for more than three persons in the household.

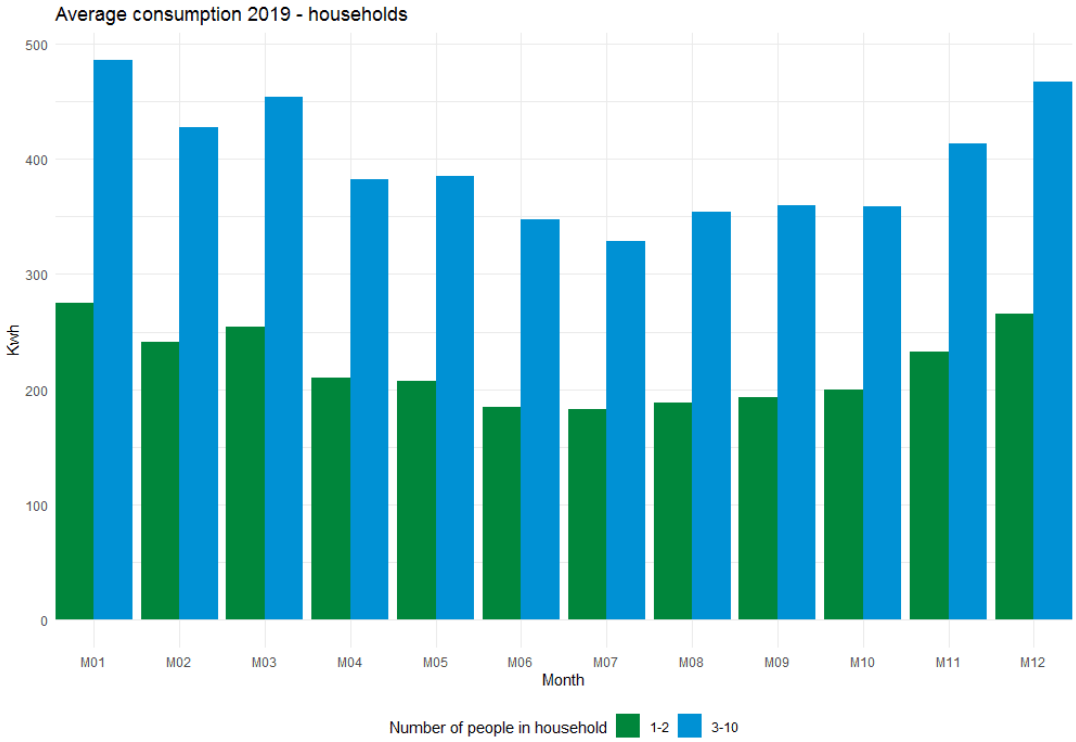


Figure 14: Average monthly consumption of Danish households

11.3 Vacant dwellings (Norwegian example)

This subsection will provide a statement of the vacant dwellings problem followed by a Norwegian case study outlining the potential of electricity consumption for identifying vacant dwellings.

11.3.1 Problem statement

The current dwelling statistics in Norway give information on the number of dwellings and on net change in the number of dwellings. Its main source is the Norwegian Real Estates, Addresses and Buildings register (REABR). It would be desirable to separate dwellings into the categories occupied dwelling, dwellings for seasonal or secondary use and vacant dwelling in accordance with the recommendations on "Occupancy status of conventional dwellings" by United Nations Economic Commission for Europe[6]. WP D has a special emphasis on vacant dwellings.

Occupancy status of dwellings at a reference period
According to international recommendations for censuses (United Nations Economic Commission for Europe 2015), the variable occupancy status of dwellings (point 905 in the recommendations) has two categories at the first level: occupied dwellings and dwellings not being occupied. The latter is at the second level divided into the two categories dwellings reserved for seasonal or secondary use, and vacant dwellings. The definition of the categories is as follows:

- occupied dwellings are dwellings used as the usual residence for one or more persons.
- dwellings not being occupied:
  - dwellings reserved for seasonal or secondary use means not main residence, and the use is either only in some seasons, or as a second home (in all seasons).
  - vacant dwellings are dwellings being vacant due to being for sale, for rent, for demolition, other reasons, or unknown reasons.

The occupancy status of dwellings can change and be dissimilar for different reference times (the period for which statistics will be published). Since Norway has only monthly aggregated electricity consumption data, we will define the reference time as a month rather than a day.

Opportunities for consumption-based separation between the occupancy status categories
Electricity consumption gives an indication of whether a dwelling is in use or not. First, consider the situation where consumption data have finer granularity than the reference period. An hypothetical example is hourly data and reference period on a certain day. Then, both the total consumption for this day as well as the hourly pattern for this day is relevant: a dwelling in use has usually larger consumption than a dwelling not in use. Further, for a dwelling in use, the hourly consumption will usually give sign-of-life: peaks will occur in a more unstructured way than when a smart meter is not in use, corresponding to switching on and off various electrical equipment. On the other hand, for a dwelling not in use, the variation during the day will typically be limited to one or

several thermostats maintaining a certain minimum temperature in the dwelling.

Contrary to the above situation, assume that the granularity of the data is equal to the reference period, e.g. daily data or monthly data for reference day and month respectively. Then the only available consumption pattern during the reference period is the total consumption level. However, assuming that the reference period is month, we will later see that the study of other monthly totals can also give relevant information on the situation at the reference month[7].

The anticipated electricity consumption level at the reference period (day or month) associated with the different occupancy status categories are:

- Occupied dwellings: Neither low nor very low consumption level except in the case of occasionally being temporary not in use at the reference month due to holiday. This means that the holiday last so long that the user chooses to turn down the electricity to a very low level. If the holiday covers the whole month, the consumption can be very low.
- Dwellings reserved for seasonal or secondary use:
  - Temporarily in use: not very low consumption, but it can range from low to rather high depending on how often the dwelling is used. Secondary use-dwellings can e.g. be used all weekdays by weekly commuters, or only on weekends as a kind of cabin.
  - Not in use: very low consumption unless the owner keeps normal heating while the dwelling is empty.
- Vacant dwellings: constantly or temporary very low consumption unless the owner keeps normal heating while the dwelling is empty. A not very low energy consumption can be also the case, if the dwelling is on sale for e.g. only a part of a month.

Elhub potential for identification of vacant dwellings
The current register sources for the dwelling statistics are insufficient for doing the separation of dwellings by occupancy status, but Elhub (Norwegian Smart Meter hub) could turn out to be a useful source in the process towards future statistics on vacant dwellings, by improving dwelling registration in REABR and by using sign-of-life data to help with identification of occupancy category (through a prediction-driven, or machine learning approach). We present, in this part of the report, a case study that investigates the potential of Elhub for identifying vacant dwellings, given the granularity of the data that have been accessible for the project and also given the possibilities of linking Elhub to the Norwegian base registers.

11.3.2 Case study set-up

Elhub data
For the case study, Elhub data were available on monthly aggregated energy consumption/production, smart meter characteristics (id, physical address, NACE, type), end user information (invoice address, id) as well as other variables that are described in

"Introducing geodata into the administrative register" [8]. The data available in time to be included in the case study data set were from June 2019 to February 2020.

Linkability
In Elhub, two types of addresses are available: physical address showing the smart meter location, and the end user address. Both address types can differ from electricity address that defines the part of the building (also mentioned as utility unit in the text) the smart meter is serving. The reasons are not only different meanings of the addresses, but also registration requirements for addresses in Elhub. For both address types in Elhub it is not obligatory to register at the utility unit level. It is sufficient to register at the less detailed level of entrance into the building, and this is the most common case in Elhub. Address at entrance level is used by the post system of Norway and thus it is sufficient for the purpose of sending the invoice. For a repairman to identify a smart meter, there is an additional free-text variable which describes how to find the smart meter after entering the building. This variable is, however, not useful for linkage.

Since addresses of Elhub mostly are at the entrance level, it is usually not possible to uniquely link a smart meter to the utility unit in REABR except for dwellings having a separate entrance into the building. For all other utility units, it is thus not possible to identify the REABR-utility unit variables that correspond to the smart meter. Utility unit variables such as area could otherwise have been used in the classification of vacant dwellings: a smaller dwelling needs small consumption whereas a large dwelling with the same small consumption could be an indication of a vacant dwelling. All entrance-level variables (for example, building-year-of-construction) of REABR can be used for the classification of vacant dwellings.

Selection of data
As the reference time for which we want to identify vacant dwellings, we have chosen February 2020. The number of smart meters available in February 2020 is around 3.29 million.

For solving the problem, we have preferred to start with a smaller set of data which would give some satisfying results rather than using all data and increase the complexity of the classification process.

Since not all smart meters were yet registered in June 2019, we have decided to use, for the case study, only those being present in all months from June 2019 to our reference time February 2020.

In the WP D case study from Norway on vacant dwellings, we exclude all smart meters being of production-type since they cannot tell us anything about occupancy status. Smart meters of combined type will be also excluded, since what they deliver to Elhub is the net consumption (difference value between monthly production and consumption). Thus, combined smart meters are systematically underreporting the total consumption,which for solar energy production is especially in the summer. Thus, the combined smart meters would complicate the classification of vacant dwellings. Getting information on the consumption part of combined smart meters is requiring additional pre-processing. Classification of occupancy status for combined smart meters will be left for future work.

Due to the linkage limitations between the REABR and Elhub, we use the Elhub data for preparing the data set for the case study. Further, we only concentrate on smart meters classified in Elhub as households (defined from the Elhub variable NACE-code, specifying the purpose of the electricity use), since only these smart meters can be vacant dwellings. In some cases, according to the NACE-code, the smart meter is registered as a household but at the same time the end user ID is a business ID rather than an ID of a person. We have decided to exclude these smart meters too, since there might be quality issues with the NACE-code. As an alternative to this exclusion, we could have chosen the classification for business/household consumption proposed in the previous ESSnet Big Data 1 WP3[7]. The number of smart meters after the exclusion is around 2,17 million.

After the linking (at the entrance level), we have information available on the number of smart meters and the number of utility units located at the same entrance-level address. The number of smart meters is greater, smaller or equal to the number of utility units registered at the same entrance-level address, and we restrict our attention to the cases of equal numbers of smart meters and utility units. In the other cases, it is less likely that each smart meter corresponds to one and only one dwelling, which would complicate the classification of vacant dwellings.

In the data set there might still be smart meters corresponding to utility units that are not dwellings, for example cellars or garages. To exclude these cases, we choose to concentrate on smart meters having a unique entrance-level address and where only one utility unit is registered on this entrance-level address according to the REABR. Our case study data set consists of around 0.89 million smart meters. In Figure 15 we get an impression of the distribution of electricity consumption for the smart meters in our data set.

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    0.0   793.7  1361.8  1448.5  2044.0  3972.4
```

Figure 15: Descriptive statistics of the electricity consumption at smart meter level on February 2020 for the case study dat aset. kWh.

11.3.3 Methodology and application

In ESSnet BD1 WP3, the possibility of identifying vacant dwellings was mostly based on hourly data, while Statistics Norway gets only monthly aggregated data. In ESSnet Big Data 1 WP3, some of the methods for identification of vacant dwelling were offered based on monthly aggregated data, and these methods will be tested in our case study. Additionally, we will show the result of the method which gave the better result in our case.

In the case study we are using the R programming language, version 3.6.0.

Unsupervised vs supervised learning
If we had a variable containing the completely or partly true vacancy type for dwellings, we could construct the classification model that minimises the classification error rate,

an approach known in machine learning terminology as supervised or semi-supervised learning, respectively. However, we have no such variable available and will perform unsupervised learning by finding natural clusters. We are aiming to split all the smart meters (dwellings) into clusters to define those clusters that correspond to vacant dwellings.

Method 1: Splitting data sets by total energy consumption level

Method 1 implies that total energy consumption for a reference time will not exceed a defined consumption level for vacant dwellings. In ESSnet Big Data 1 WP3, the value zero was used as a limiting consumption level. Applying the limit in our case we get that:

- There are 0.81% cases where total consumption in February 2020 does not exceed a zero limit.

By using only consumption from the reference time, we can only get information on low/high energy consumption, which does not describe vacancy. Having low energy consumption during the reference month does not necessarily mean that that the dwelling is vacant. It can also be occupied, in seasonal use or in secondary use, since all these situations are consistent with being not in use during the reference month. Thus, we should use more months than just the reference month to identify vacancy. In this case we get that:

- There are 0.41% cases where total consumption in the period June 2019 - February 2020 does not exceed the zero limit.

We can choose to increase the limit value so that a vacant dwelling would correspond not only to zero but also to a very low electricity consumption. For example, we can use a density plot to identify the limit. In Figure 16 we see that density of energy consumption seems to be a mixture of three distributions corresponding to each of three groups. The distribution to the right is for dwellings in use during the whole reference month, the middle one is probably for dwellings being partly in use such as secondary use dwellings and seasonal use dwellings. The distribution to the left is probably for vacant dwellings. We notice that these three distributions are overlapping, so when we try to separate the three groups (using the consumption values of the local density minimum), we cannot separate perfectly: at the lower minimum separating between vacant dwellings and secondary/seasonal dwellings, we see that some vacant dwellings will end up in the secondary/seasonal dwelling group and vice versa. This also corresponds to our earlier comment that not only vacant dwellings can be "not in use" during the whole reference month.
If we compare the situation for February in Figure 16 with other months, we see in Figure 17 that the local minimum consumption value splitting the left groups and the middle groups, is approximately equal for all months.

As a limit we can choose to use the very left bin in Figure 16 (values between in interval: [0;10]) and using the same limit for all months. Then we get:

- For 0.63% of the cases the total consumption in the period June 2019 - February 2020 does not exceed the 10 kWh limit.

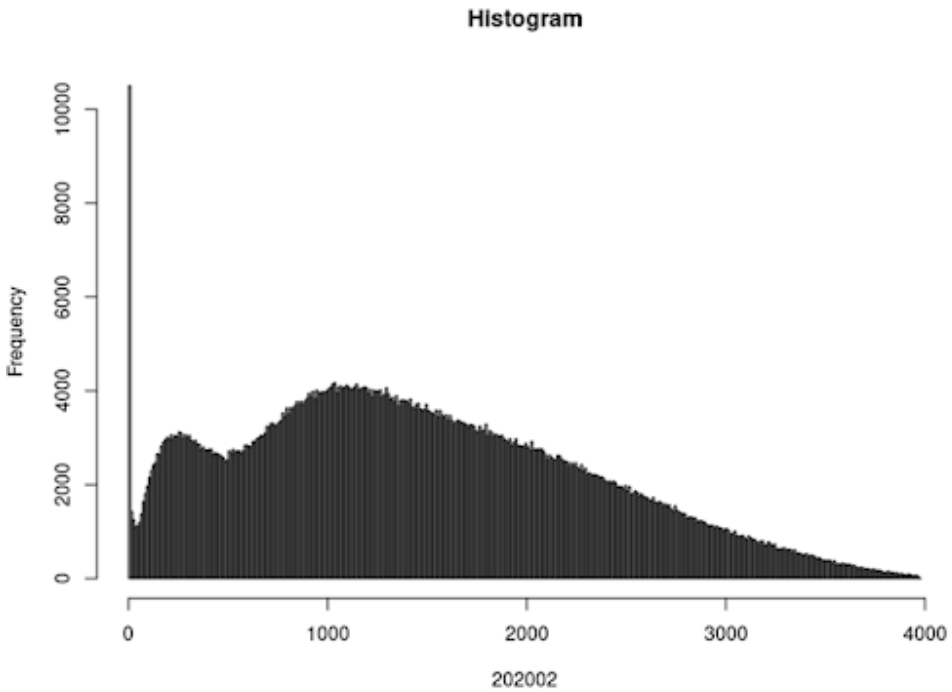


Figure 16: Histogram of energy consumption for February 2020

If we use this method to classify into vacant dwellings versus seasonally/secondary use for a reference month, we do the following errors due to the overlapping distributions: we either fail to include vacant dwellings (having consumption above the limit), or we fail to include occupied or seasonally-occupied dwellings (having consumption below the limit).

Method 2: k-means clustering applied to all months together

In ESSnet Big Data 1 WP3, Estonia performed an application of unsupervised machine learning on monthly energy consumption data to identify consumption pattern clusters[7]. In our case study, we will reproduce the method.
As a first step, we apply data normalisation : $Z = \dfrac{x - mean(x)}{stdev(x)}$. The normalisation was applied separately for each month.

Further, the elbow method will be applied to identify the number of clusters to be used in k-means clustering (see Figure 18 for our situation when we use only the monthly consumption values and Euclidean distance). In the figure, on the x-axis (k_opt) is the chosen number of clusters and on y-axis is the inertia. Inertia is the within-cluster sum of squared errors, where the error is defined as the distance between a point in a cluster and the cluster centroid. The smaller the inertia, the closer are the points in the cluster to the centroid and consequently also to each other. In other words, the smaller the inertia is, the denser is the cluster.

To automatically find an optimal number of clusters, we draw a straight line from the very left point of the curve to the very right point and calculate the distance from the straight line to each point on the curve (k_opt, inertia). The point on the curve having the maximum distance to the straight line is chosen. From this method, the optimal

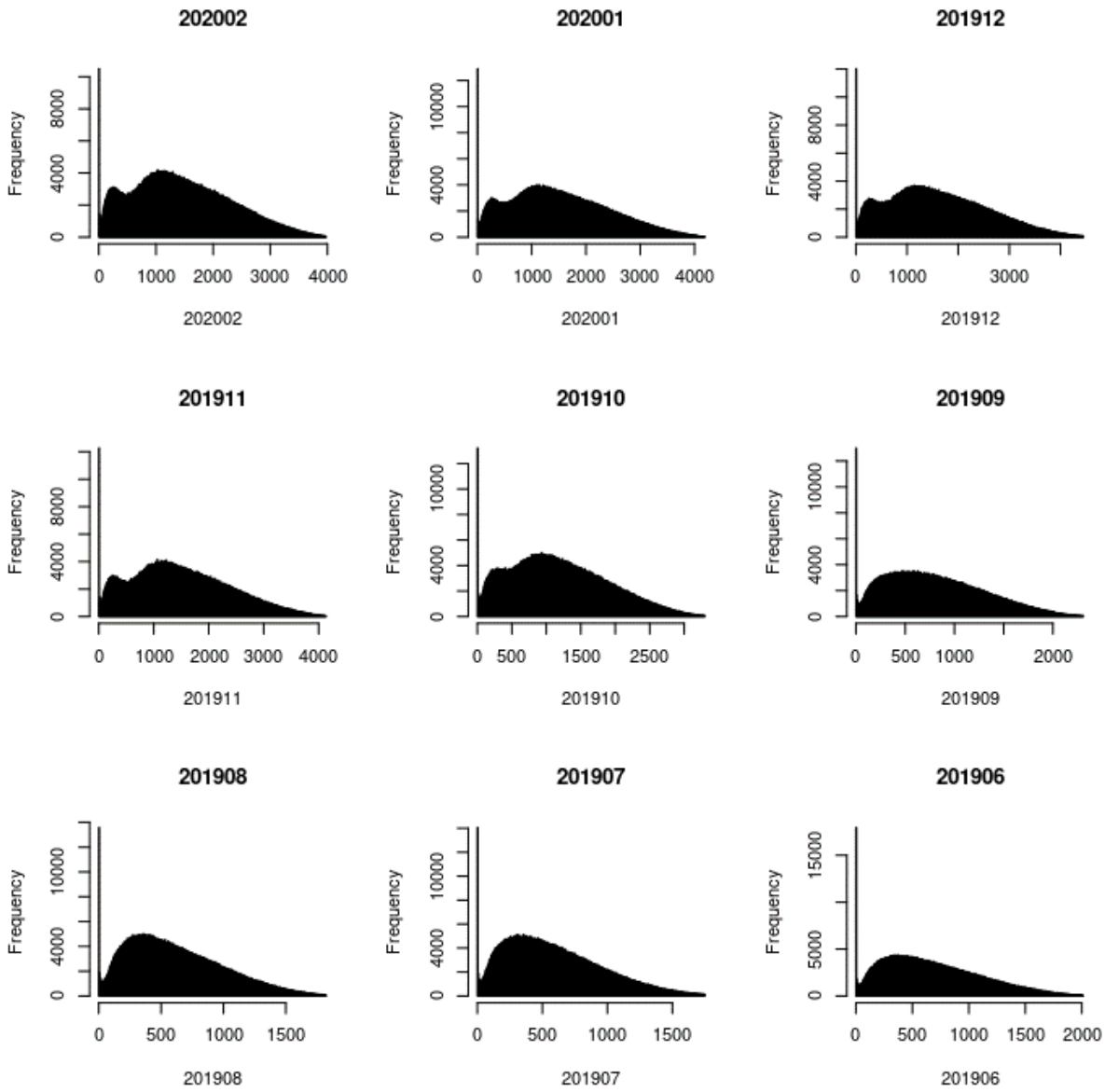


Figure 17: Histogram of energy consumption for June 2019 - February 2020

number of points is 8.

Since the number of optimal nodes is 8, we use k=8 in the k-means method with Euclidean distance, and we get the cluster centres in Figure 19. As we see, the k-means method applied on the complete data set for only energy consumption variables did not really manage to split consumption by patterns. Increasing the number of clusters up to 16 or 30 did not give positive results.

The reasons that k-means with Euclidean distance was struggling to make a split by patterns can be illustrated by a simple example. The points (0.3, 0, 0.4), (0, 0.5, 0) are equally far from (0, 0, 0) so that the two former points can appear in the same cluster. To solve the problem, we will describe a modified method (Method 3) that combines the k-means method with Method 1. Other approaches could be to adjust the k-means method by, for example, significantly increase the number of clusters, or use feature engineering

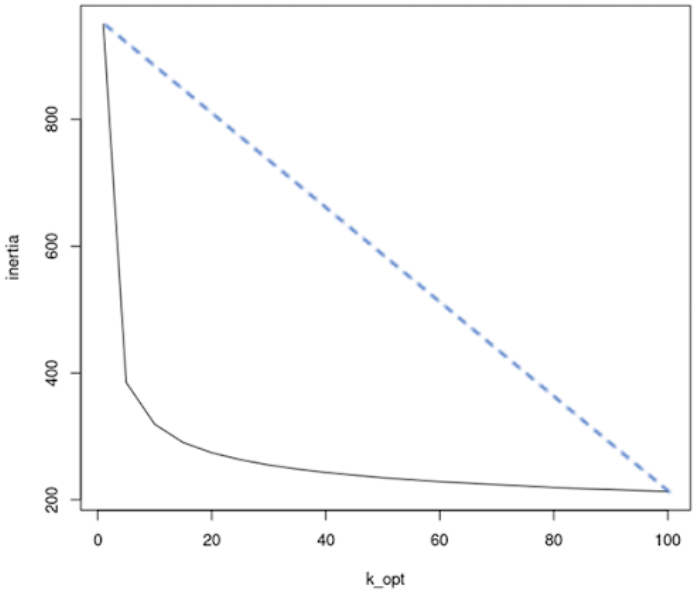


Figure 18: Elbow method for k-means clustering

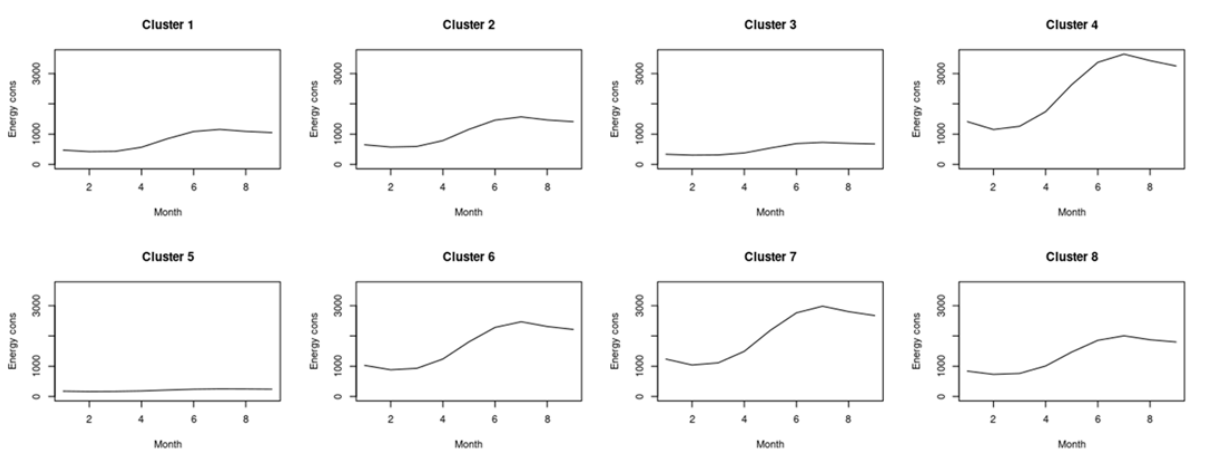


Figure 19: Results of k-mean clustering with 8 cluster centres

to construct new variables which would be significant for capturing pattern shape: for example the difference between the consumption at the reference time of the current year compared to the previous year, the variance, the maximum value, the second minimum value, or other.

Method 3: combining Method 1 and k-means clustering applied to each month separately

Method 3 is a combination of Method 1 and Method 2, and where a data preprocessing step was not required.

Step 1: classification of smart meters into "probably in use" or "probably not in use" at the reference month.

Cluster	Min	Mean	Max
1	0	1.55	12.7
2	12.8	24.0	36.2
3	36.2	48.3	60.0
6	60	1476	3972

Table 2: Average, minimum and maximum monthly consumption in the four groups of February 2020 using the $k_i + 1$ -split method

At the first step, we want to split all smart meters in two groups: those in use and those probably not in use, by using limit values which were described in Method 1. For each month, we calculate the leftmost local minimum on the density plot of the month. For constructing the density plot, we are using a histogram with bins of size 20 and for each bin finding whether the frequency value in the previous and the next bin is higher than the current bin. We take as the limit the right value of the bin interval which was defined to be a local minimum.

The maximum of the limit values for all observed months is equal to 40kWh. Since a consumption of 40KWh per month is very low (on average in Norway, a household is consuming 45.5 kWh per day[9]), the cluster (for a certain month) having lower value than the limit value can be classified as most probably not in use.

Due to overlapping distributions (see the earlier explanation for Figure 16), some of the not-in-use dwellings have a consumption higher than the local minimum-limit above. To include some of these dwellings, we choose for each month to include dwellings whose consumption exceeds 1.5*limit. This unfortunately means that we also include some in-use dwellings, but our approach is to identify these dwellings in Step 2.

Step 2: separating "probably not in use"-dwellings by patterns

For each month separately: For consumption values which are lower than 1.5*limit value, we apply k-means clustering. Firstly, we run the elbow method (described under Method 2 above) to identify what number k of clusters should be used as input to k-means clustering. Then for each month i, we have $k_i + 1$ groups, where k_i is the number of clusters identified by the elbow method, and the extra cluster consists of the smart meters that during the previous step have been classified as most probably in use. If the elbow method applied to a month's data resulted in $k = 3$, we have four groups for this month. We denote this approach as the $k_i + 1$ -split method.

For February 2020 there are 16 609 units with energy consumption lower than 60kWh (1.5 * limit). Using the k-means method for February 2020, the data are split into four groups for which averages, minimum and maximum can be found in Table 2 (the fourth group is denoted as Cluster 6).

After running the $k_i + 1$ -split method for each of the nine months, each smart meter belongs to nine clusters forming a sequence of monthly clusters. As an example, a smart meter may belong to 213 663 113, where the first digit is the monthly cluster of February 2020 and the last is that of June 2019. We will now consider these 5711 new clusters, and

since we want to define vacant dwellings for February 2020, we exclude those units that in February were classified as most probably in use (group 6), thus we get 3340 clusters.

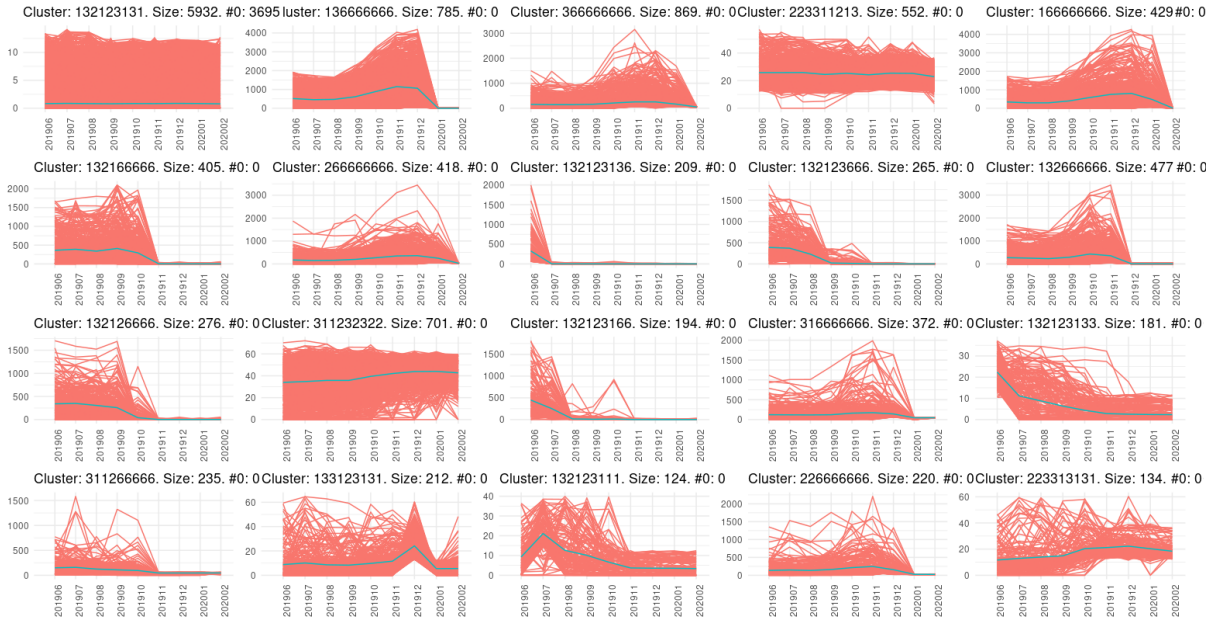


Figure 20: Energy consumption (red lines) and cluster centres (blue lines) for 20 the biggest clusters. (In captions to clusters: 'Cluster'- the cluster name, 'Size' - the cluster size and '#0'- a number of smart meters having zero consumption in the observed period.)

Figure 20 shows the energy consumption and cluster centres for the 20 largest clusters (containing the largest number of smart meters). We see the different consumption patterns and notice that cluster "132 123 131" (1st row, 1st column) has constantly low energy consumption, and thus the units from the cluster can be classified as vacant dwellings. We get:

- 0.66% cases in cluster "132 123 131" classified as vacant.

For the rest of the clusters, additional analyses are needed to make the classification. For example, we see that cluster "136 666 666" (first row, second column) might contain all occupancy types, since it had a high energy consumption all months excluding January and February 2020. Two examples are an occupied or secondary use dwelling where the residents are on holiday these two months, and a seasonally used dwelling that is only used June-December.

At this step, the number of clusters that we get is high, and lots of clusters contain just a few units. In Step 3 below we want to add each small cluster to the most appropriate of the bigger clusters, to reduce the number of clusters. We will by *representative clusters* refer to these bigger clusters.

Step 3: combining small clusters with representative clusters

A challenge is to determine the number of clusters to choose as representative. Currently, it was decided to use a threshold for cluster size. By using the threshold as a lower bound value for cluster size, all small clusters will be excluded. However, more investigation is

needed concerning which method to use for choosing the representative clusters or understanding small clusters.

A threshold of 30 units was chosen. Thus, all clusters containing at least 30 units were chosen to be representative. This leads to 42 representative clusters containing 10361 of the units, whereas each of the 6248 remaining clusters should be added to one of the representative clusters.

Figure 21: Energy consumption (red lines) and cluster centres (blue lines) for the 20 biggest clusters after application of Random Forest for the smaller classes

The smart meters that belong to the representative clusters were chosen for training a supervised machine learning model. As a test, we used weighted RandomForest with ntrees = 500, with no train/validation split and with the following independent variables: the consumption range of a smart meter for the observation period (nine months), the max consumption during the same period, the max difference between two consecutive months during the same period, the variance of the consumption values for the period, and consumption values for each of the last four months in the period. The weights to the variables were assigned correspondingly: the highest weight for the earliest independent variable mentioned and the lowest to the latest. One can notice that splitting into training and validation data sets was omitted as the requirement of doing this split to avoid overfitting sometimes is considered not so strict for random forest compared to many other methods, although it is highly recommended. Nevertheless, the primary goal here was to find a base solution as a starting point, and then to make further improvements of the method based on the random forest output. However, when choosing optimal parameters and comparing different methods/models, one should not neglect the splitting. The method presented here does not claim to be the final choice, finding a better one is still needed.

In Figure 21, we see the result of using the RandomForest method to add the 6248 remaining units to the representative clusters. Most of the clusters kept their pattern from

Figure 20, while other clusters show that some improvements of the classification method is still needed for some of the 6248 units.

Model evaluation

In our case we did not have any variables available to run supervised machine learning or to directly evaluate the vacancy results from the unsupervised models. To evaluate a model, we chose to check if smart meters with zero consumption in all months belong to the same cluster (Figure 20). We also looked at descriptive statistics in the cluster, where the range and max value should not be far from zero (Table 2 and Figure 22). Further we investigated the pattern of energy consumption (Figure 20 or Figure 21).

```
    202002          202001          201912
Min.   :  0.0   Min.   :   0.0   Min.   :   0.0
1st Qu.:166.6   1st Qu.: 185.0   1st Qu.: 184.4
Median :297.4   Median : 324.3   Median : 335.3
Mean   :296.9   Mean   : 336.4   Mean   : 373.0
3rd Qu.:434.7   3rd Qu.: 479.4   3rd Qu.: 512.4
Max.   :591.7   Max.   :2987.3   Max.   :3415.0
```

Figure 22: An example of a descriptive statistics for December 2019 - February 2020 for a cluster containing zero consumption for all months and with range and max values for February 2020 (month of interest) far from zero. The conclusion is that the cluster is far from containing only vacant dwellings and, consequently, the clustering method should be improved. The example is not obtained from Methods 1-3 described here.

Usage of end user id

As an example of how the use the change of end user id, we consider cluster "136 666 666" (1st row, 2nd column) in Figure 21. The size of the cluster is 785. The pattern is showing that in the last two months, there was a valuable reduction in energy consumption. By looking at user ID, we notice that for 155 smart meters, there were changes in smart meter owners either from December to January or from January to February. This could mean that these apartments were for sale in February or that the new owner did not move into the apartment in January or February. In both cases, this means that the unit can be classified as unoccupied.

Other methods which were tested but did not give positive result

Among the popular cluster methods, we also tested DBSCAN and hierarchical clustering which however was not possible to perform on the whole data set since the R-session crashed due to lack of memory. These clustering methods were then performed on just a sample from the dataset, with the idea of applying supervised machine learning techniques to the rest of the smart meters that were not used for clustering. However, this did not lead to clusters with clear patterns.

Using information on building characteristics available from REABR together with energy information as input to the clustering model was also attempted in this case study.

All variables were normalised and combinations of different weights between variables were tested. There were no substantial improvements in the clustering, in some cases this even lead to worse patterns in clusters.

Feature engineering based on energy consumption variables were also tested. However, we did not find new variables that would be significant for capturing pattern shape.

11.3.4 Summary

In Norway, to implement official statistics on vacant dwelling based on the administrative registers by themselves is not possible since there is not enough information available. Using Elhub (the smart meter hub) by itself to produce vacant dwellings statistics requires knowledge on the type of utility unit being served by the smart meter. However, this information is not known from the Elhub data. To extract the necessary information, linking of Elhub and REABR was attempted.

Linking Elhub and REABR on utility unit level was not possible for many of the smart meters because there is no electricity address (address of the utility unit that the smart meter is serving) specified in Elhub, and neither the physical nor invoice address are necessarily coinciding with the electricity address. Linking on entrance-level provides in many cases different number of smart meters and utility units in a building, and then it is challenging to determine whether a smart meter is serving a dwelling.

As far as the classification part is concerned, there are practically no dwellings in REABR that with certainty can be declared as vacant based on the administrative registers. Thus, an unsupervised learning approach was used for identifying vacant dwellings, and then the estimation of the classification error becomes complicated.

From this case study, clustering of smart meters into the three occupancy groups (occupied, seasonally/secondary use, vacant) by only using information on monthly aggregated energy consumption is not possible. It is neither possible to adequately separate all the vacant dwellings from the non-vacant dwellings: some smart meters have an energy consumption pattern over months that might correspond to both a vacant and a non-vacant dwelling. Smart meters with permanently low energy consumption were defined as vacant, together with those smart meters with an end user change and suddenly reduced energy consumption by the time of the reference period.

Since smart meters are not installed in all utility units, e.g. some users rejected installation, and since the occupancy groups are poorly identified by monthly aggregated energy consumption, it should be considered to combine Elhub with other data sources containing information on human activity by location, e.g. mobile phone data.

11.4 Vacant dwellings (Estonian example)

Identifying vacant or seasonally empty living spaces from electricity consumption data can be relevant for housing statistics as it provides information where people actually live. The results can be used in the population and housing census. For example Estonia has 720 387 unique metering points. The consumption histogram of those metering points can be seen in Figure 23.

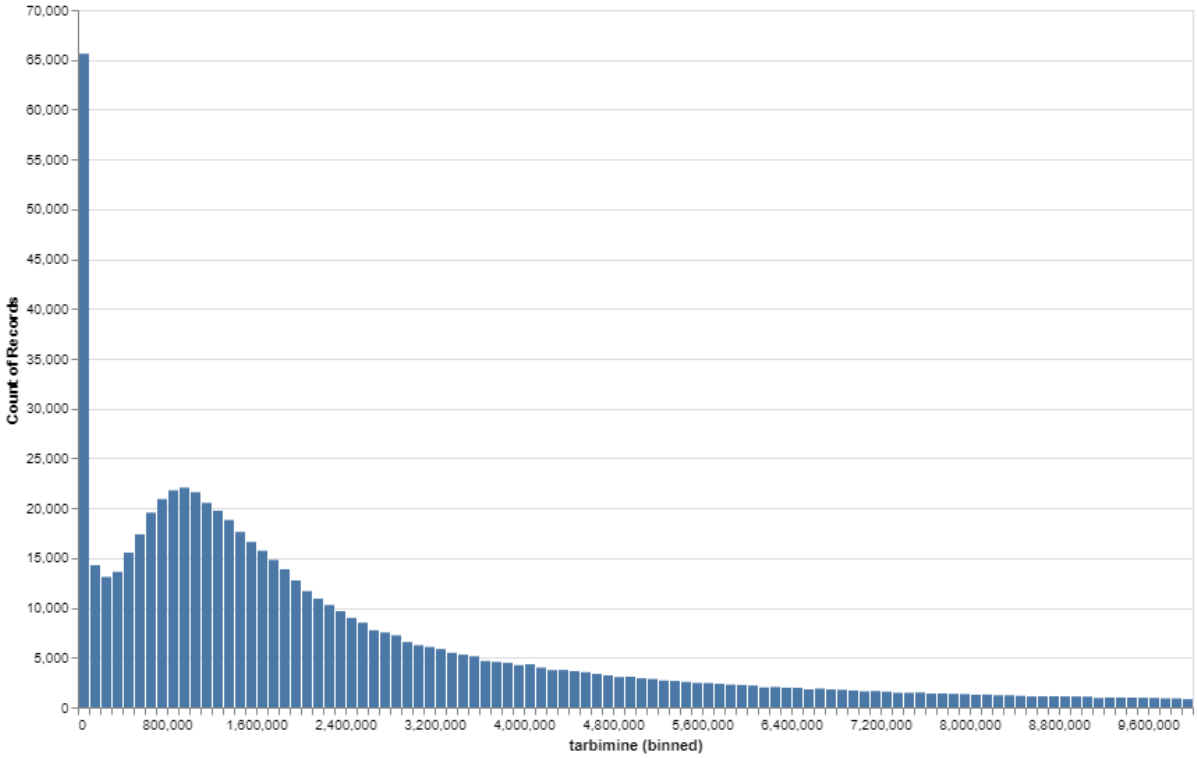


Figure 23: Histogram of Estonians energy consumption in 2019

One way to find out which of these metering points are empty (vacant), seasonal (summer), and occupied houses is to use clustering. Cluster analysis could give valuable insights from our data by grouping the data points with a clustering algorithm. Since the data are unlabelled, the only way to cluster the data is to apply unsupervised machine learning. The most used clustering algorithms are,

- K-means,
- Hierarchical clustering,
- Mixture model,
- DBSCAN,
- etc.

Out of all these clustering algorithms, we chose K-means as it is the most prevalent algorithm used for clustering smart meter electricity consumption data[10]. The next decision is about the distance metric. It is observed that DTW (dynamic time warping) metric works better when the data possess high granularity such as 15 minute or less. Otherwise, Euclidean metric performs better with k-means on data of hourly and daily granularity.

Since k-means requires the number of clusters to be specified, the elbow method is used to determine the number of clusters.

11.4.1 Data preprocessing

The Estonian consumption data set contains hourly data for each smart meter. For ease of analysis, the hourly granularity has been aggregated to days, resulting in a data set where each smart meter has 365 instances of consumption. Table 3 is an example from the aggregated data set where the point_ID values were populated with pseudo ID-s to ensure anonymity.

point_ID	Day	Consumption
12 345	2019-02-04	237
12 345	2019-03-06	843
12 345	2019-06-20	932
	...	
21 345	2019-07-19	699
21 345	2019-08-26	795

Table 3: Example of Estonians aggregated consumption data set

Part of the data preprocessing was to convert the Day column in Table 3 from year-month-date format into day-of-year format. The table is then transposed from a long data set into a wide data set, so that each unique point_ID has one row with 365 columns that hold the consumption data for the year. All missing and Not a Number (NaN) values are imputed with 0. Table 4 shows the data prepared for clustering.

point_ID	1	2	3	...	363	364	365
12 345	427.0	427.0	427.0	...	458.0	438.0	424.0
12 445	1053.0	968.0	810.0	...	775.0	1885.0	3505.0
12 545	913.0	836.0	774.0	...	2116.0	1686.0	997.0
12 645	1775.0	1398.0	1453.0	...	1404.0	1456.0	1389.0
12 745	129.0	1785.0	234.0	...	472.0	2699.0	896.0

Table 4: Example data set that is prepared for clustering.

11.4.2 Methodology

The clustering process can be divided into five sections.

1. Split the data set into eight subsets based on consumption threshold from the histogram (Figure 23).
2. Run the elbow method separately for the eight subsets and look for probable cluster numbers.
3. Perform k-means clustering in those eight subsets with number of clusters decided from the previous step.

4. Analysis of the result.

5. Validate the clusters and the result.

Splitting the data set into smaller subsets gives more control over the process and makes it easier to identify different clusters. For example, the first bin in Figure 23 contains around 65 000 metering points where the yearly consumption is between 0-130 kWh. The second bin contains around 14 000 metering points where yearly consumption is between 130 - 260 kWh. It is logical to start looking for empty dwellings in those consumption regions.

11.4.3 Results

After clustering there where 11 distinct groups that shared similar electricity consumption patterns (Table 5).

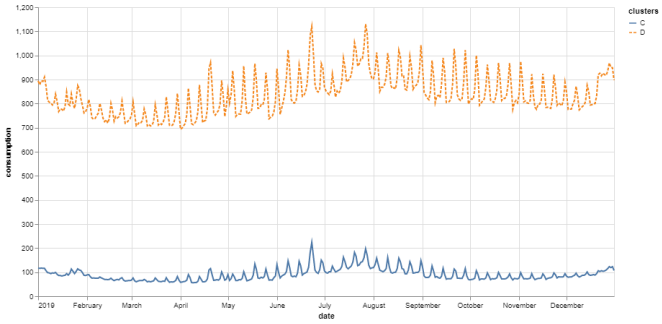
Cluster	**metering points**
vacant	105 783
occupied_low	110 414
summer_house	20 288
occupied_1	94 948
occupied_2	262 988
occupied_spring	15 588
occupied_autumn	15 039
industry_1	66
industry_2	95 752
industry_3	1
outlier	4

Table 5: Results of clustering and number of metering points in each cluster

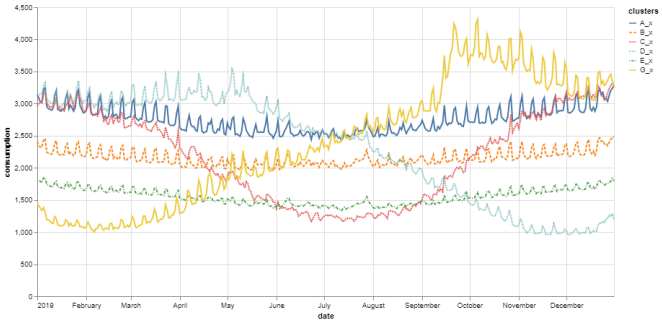
There where 105 783 metering points where the daily average consumption of the cluster was lower than 1 kWh (Figure 24a). Estonia's largest energy provider has defined a 250 kWh yearly consumption threshold for vacant living spaces, hence this cluster can be classified as vacant.

We observe 20 288 unique metering points where the consumption increases during the summer months and there are high consumption peaks during the weekends (Figure 24g). This cluster was classified as summer houses. It is necessary to point out that there where no summer houses that exceeded yearly consumption of 4500 kWh.

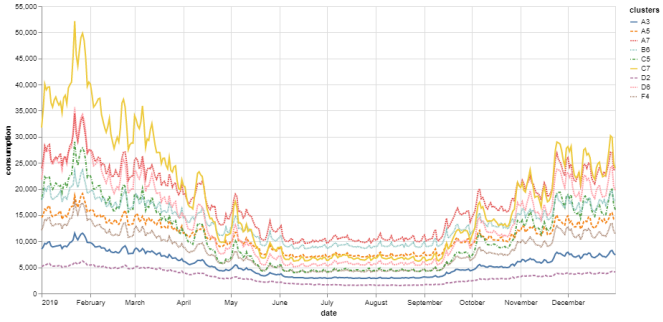
It was possible to identify five different consumption patters for occupied living spaces (Figure 24b - 24f). The occupied_low cluster consists of metering points where the daily average consumption was between 1.5 - 3 kWh. The daily average consumption for this cluster is low, but clear consumption patterns indicate occupancy of the households.(Figure 24b). The highest number of metering points (262 988) is in cluster occupied_2 (Figure 24d) where the daily energy consumption throughout the year is quite constant. We see a small decrease during the summer months but it is significantly



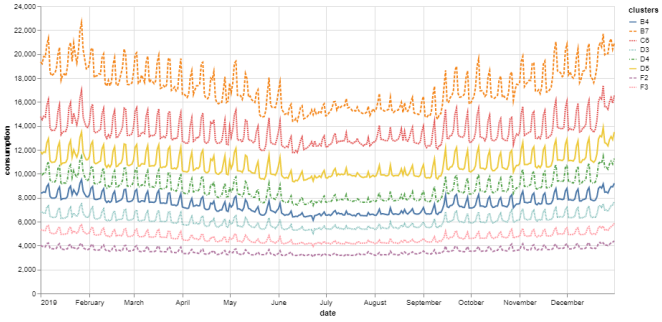
(a) vacant cluster



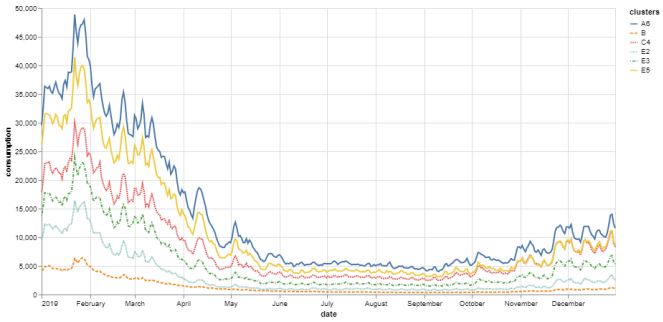
(b) occupied_low cluster



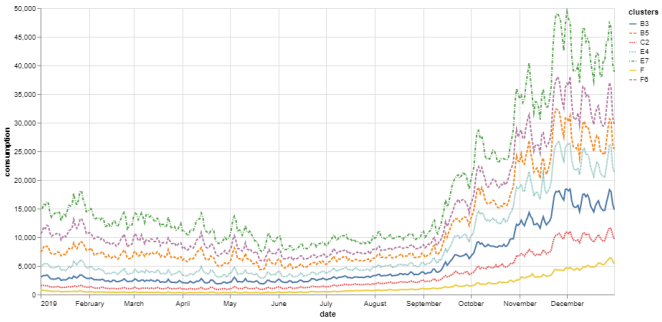
(c) occupied_1 cluster



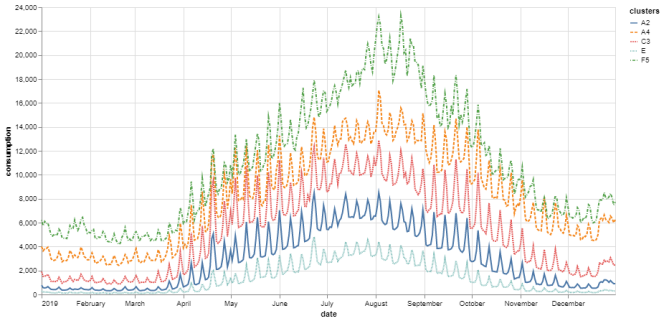
(d) occupied_2 cluster



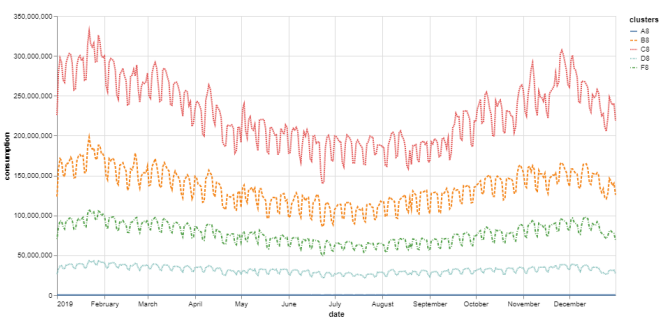
(e) occupied_spring cluster



(f) occupied_autumn cluster



(g) summer_houses cluster



(h) industry_2 cluster

Figure 24: Average consumption patterns of different clusters

smaller than in occupied_1 cluster (Figure 24c).

In Figure 24h we see consumption patterns for metering points that where classified as industry. Compared to households the average daily consumption for industrial metering points is higher, starting from 45 000 kWh a day, and it is quite constant throughout the year. It is also possible to see a drop of consumption during the weekends that clearly distinguishes this cluster from households.

12 Conclusion

Smart meters provide a possibility to innovate existing statistics and produce new statistics. Development of smart meter hubs provides a possibility to ease transfer of the data by getting them from a centralised place.

This work was devoted to establishing official statistics production based on smart meter data. Below, a high-level outline of the recommended production process is provided:

- Before receiving the data, explore metadata and conceptualise the data you need.
- Come to a detailed agreement with data owners, specifying a data exchange protocol, file structure, and file content.
  - Files containing Smart meter data will be large and some information (background information) does not change often. To optimise the receiving process it is beneficial to split the data into several files: consumption/production data, background information on the end user and on smart meters.
  - Aggregation level of consumption/production data by time period should be considered based on needs. Hourly data lead to increased size of the data files and are not always available and not always necessary. At the same time, monthly aggregated data might not be enough to satisfy all needs. As an example, in this report, one can find business and household statistics for monthly consumption which can be obtained based on monthly aggregated data. On the other hand, daily data give better possibilities for identifying vacant dwellings. One could also consider averages for weekends and weekdays. The decision will be based on conclusions from the metadata exploration.
- Control the quality of the received data. Validate that data were received as agreed, that they do not contain duplicates, and make other controls as described in the section "Data validation".
- Choice of tools for data storage is strongly dependent on data size and the needs and requirements of the NSI. An evaluation of Apache Hadoop and PostgreSQ is find in "Appendix A"
- The raw data should be prepared and ready for production of statistics.
  - Smart meter data are sensitive data since they contain names and national identification numbers, which should be replaced with correspondent pseudonymous values
  - Usage of geocoding is beneficial as it will make address codes unaffected by address changes.
  - Linking smart meter data with registers allows for extracting additional information, however, might be challenging.

In this work, we provide examples of statistics on household and energy consumption. The statistics were obtained from smart meter data and are already implemented in production. In addition, it is shown that useful information can be extracted from smart meter data. With daily data, it is possible to extract businesses vs dwellings patterns.

A Danish example shows how smart meter data have been used to explore the effect of the lockdown due to the Covic-19 pandemic on energy consumption. The results on identification of vacant dwelling based on daily data will be input to the Estonian population census 2021.

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A When to use big data tools

Estonia has evaluated two solutions for data storage and processing:

1. Apache Hadoop framework, both as a data storage backend (HDFS) and data querying and processing platform (Hive and Spark).
2. PostgreSQL database together with widely used data science tools available for R and Python languages.

A.1 Apache Hadoop

The Apache Hadoop software library is a framework that allows for the distributed processing of large data sets across clusters of computers using simple programming models. It is designed to scale up from single servers to thousands of machines, each offering local computation and storage. Rather than relying on hardware to deliver high-availability, the library itself is designed to detect and handle failures at the application layer, delivering a highly-available service on top of a cluster of computers, each of which may be prone to failures.

Hadoop Distributed File System (HDFS) is a distributed file system that provides high-throughput access to application data.

Hive is a data warehouse infrastructure that provides data summarization and ad hoc querying.

Spark is a fast and general compute engine for Hadoop data. Spark provides a simple and expressive programming model that supports a wide range of applications, including ETL, machine learning, stream processing, and graph computation.

A.2 PostgreSQL, R and Python

PostgreSQL is a powerful, open source object-relational database system that uses and extends the SQL language combined with many features that safely store and scale the most complicated data workloads.

R is a programming language and free software environment for statistical computing and graphics. R has grown from a statistic package to a popular data science toolset with a thriving community, lots of add-on packages and visualisation options.

Python is a high-level, general-purpose programming language. One of the most popular programming languages in the world, Python has extensive set of data science packages.

A.3 Choosing the tools

Probably the most important aspect while choosing the tools is to look at the dataset size. Estonia receives hourly data from all metering points, which sums up to about 6.5 billion records per year (Table 6).

Timeframe	Number of records
One hour	750 000
One day	18 000 000
One month (approximately)	540 000 000
One year	6 570 000 000

Table 6: Hourly smart meter records and number of rows

In official statistics hourly and even daily data is rarely needed, so to optimise further processing, we calculate aggregates for every day, month and year and store it along with the raw hourly data.

Aggregates	Number of records
365 days	273 750 000
12 months	9 000 000
One year	750 000

Table 7: Aggregated smart meter data and the number of rows they

Using the aggregate datasets (Table 7) means that raw data almost never need to be accessed and resource requirements for memory and processing power are considerably smaller. In our case, due to the relatively low number of metering points, we have found that using PostgreSQL for data storage is a better solution compared to Hadoop. On the same hardware, response times for most of the queries are up to ten times worse with Hadoop tools.

Another thing to consider is the complexity of installing and managing Apache Hadoop cluster - this requires specific skills that might not be readily available in every organisation. Also with small scale setup more hardware resources are needed for Hadoop compared to the use of simple relational database. Apache Hadoop should be considered in cases when

- data size is very large (more than ten million metering points with hourly data)
- highly parallelizable algorithms are used that can benefit from the distributed processing

B COVID-19 indicators

Denmark followed both households and businesses during the lockdown of the Danish society. The lockdown was initiated in week 11 of March and lasted well into June. Danish employees were sent home to work, which can be seen in the increase in electricity consumption during the day in Figure 25. Many businesses experienced a significant decrease in activity, either due to lost turnover or shift in production from the workplace to the household.

B.1 Households

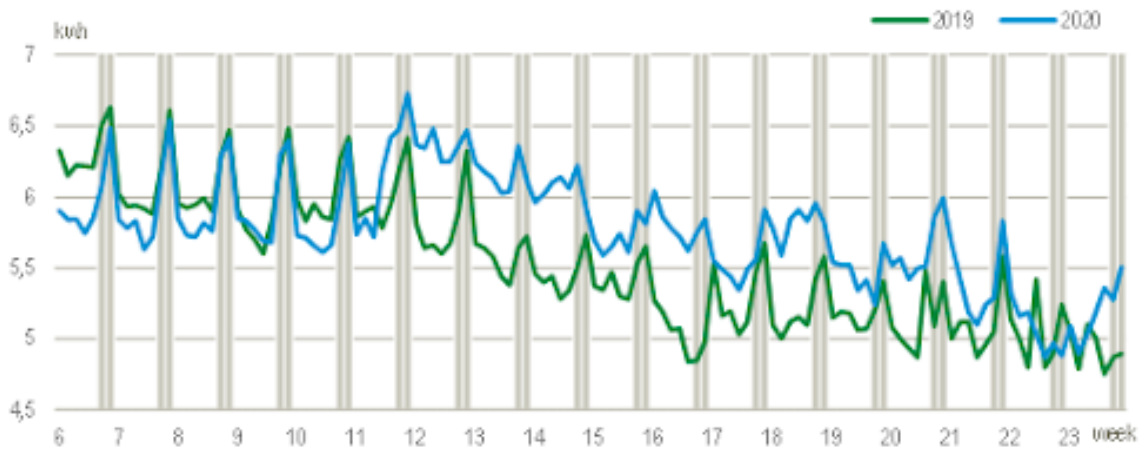
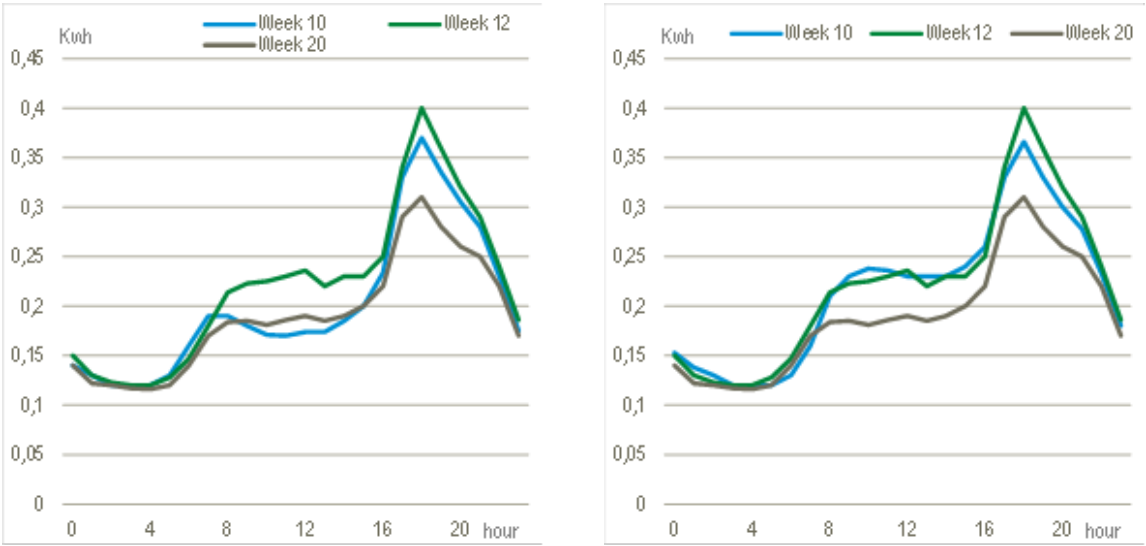


Figure 25: Median daily consumption in Danish households. Grey vertical bars: weekend.

Breaking the consumption down on households, it is evident that households normally experience a morning peak followed by a clear decrease until the evening peak, but during the lockdown households would handle work, homeschooling and daycare at home during the day. The consumption on weekdays resembles the consumption on weekends as can be seen in Figure 26a and Figure 26b.

B.2 Businesses

Businesses were found following the same methodology as in the business methodology section. Instead of linking by an addressecode, the self-reported business register number was used. The advantage of that is to find which business the metering point belongs to. The disadvantage can be that there is a difference between the business number as a payer of the invoice and the actual user on the physical address of the metering point.
This section gives an overview of the immediate reaction to the lockdown and the following reopening for a selection of sectors. Figure 27 shows the weekly median consumption of businesses with index in week 2 2020. The figure shows only six sectors. The rest of the sectors can be followed at **https://www.dst.dk/da/Statistik/eksperimentel-statistik-covid-19**.



(a) (b)

Figure 26: **a** - daily consumption on weekends; **b** - daily consumption on weekdays

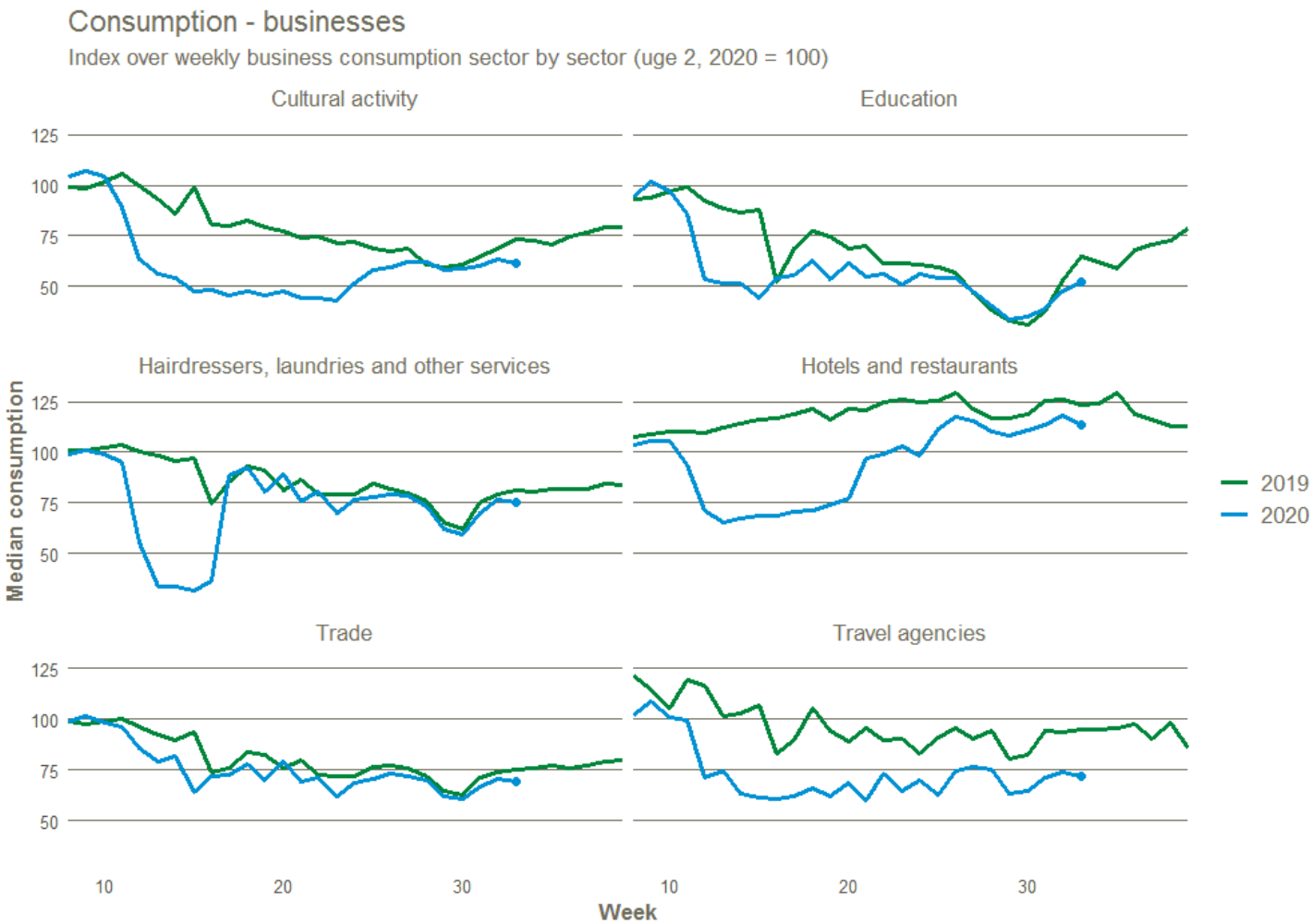


Figure 27: Danish businesses electricity consumption during COVID-19 lockdown